

ONSSA PROJECT

Chapter 1 — Executive Summary

1. Executive Summary

The National Office for Food Safety (ONSSA) manages and publishes a large body of regulatory information covering food safety, sanitary regulations, legal texts, technical guidelines, administrative procedures, and official documentation. This knowledge base is continuously consulted by internal staff, inspectors, administrative personnel, and other stakeholders who require rapid and reliable access to official regulatory information.

Although these documents are publicly available through the existing ONSSA platform, accessing the appropriate information often requires manually navigating numerous web pages and legal documents, including laws, decrees, ministerial orders, technical guides, official bulletins, forms, and scanned documents. As the regulatory corpus continues to grow, locating precise information becomes increasingly time-consuming and complex.

The objective of this project is to develop an intelligent regulatory assistant capable of understanding, retrieving, and explaining ONSSA regulatory information exclusively from official documentation. Rather than replacing the current ONSSA platform, the proposed solution is designed as an independent Artificial Intelligence service that integrates seamlessly with the existing information system through a REST API.

The system relies on a Retrieval-Augmented Generation (RAG) architecture combined with a domain-specialized Large Language Model (LLM). The LLM is not intended to memorize the complete regulatory corpus. Instead, regulatory knowledge is retrieved dynamically from a structured knowledge base generated from official ONSSA documents. The model is specialized through parameter-efficient fine-tuning to adopt ONSSA-specific terminology, regulatory language, response structure, citation format, and instruction-following behavior, while factual information is provided at inference time through document retrieval.

The project architecture is organized around a centralized Knowledge Ingestion Pipeline, which constitutes the core of the entire system. This pipeline is responsible for discovering, collecting, processing, cleaning, structuring, and transforming heterogeneous ONSSA documents—including HTML pages, PDF documents, Word files, official bulletins, scanned documents, forms, images, and technical guides—into a unified and structured Knowledge Corpus. This corpus subsequently serves two complementary purposes: generating the Instruction Dataset required for model specialization and producing vector embeddings used by the retrieval system.

To reduce technical risk and validate the complete architecture before scaling to the full ONSSA repository, the project follows an industrial Vertical Slice development strategy. The initial implementation focuses exclusively on the **Regulation** → **Transversal Regulation** → **Food Safety** section of the ONSSA regulatory repository. This limited yet representative subset enables validation of the entire end-to-end pipeline, including document ingestion,

corpus construction, retrieval, model specialization, response generation, evaluation, deployment, and integration. Once validated, the same architecture can be extended incrementally to additional regulatory domains without requiring fundamental architectural modifications.

Given the sensitive nature of regulatory information and institutional requirements regarding confidentiality and digital sovereignty, the entire solution is designed for fully on-premises deployment. All components—including document processing, vector indexing, model inference, and API services—operate locally without relying on external cloud services or proprietary AI APIs. This approach ensures complete control over data, compliance with security requirements, and long-term maintainability.

The final deliverable is a production-ready AI service exposing a REST API that can be consumed by the existing ONSSA platform. Users interact with the platform through its current interface while AI requests are transparently routed to the specialized regulatory assistant. For every response, the system retrieves the most relevant official documents, generates a grounded answer, and returns precise references to the source documents used to produce the response. This guarantees transparency, traceability, and user confidence while significantly improving access to regulatory knowledge.

The architecture proposed in this document establishes a scalable foundation for progressively transforming the complete ONSSA documentary repository into a trusted institutional knowledge system while preserving document authority, regulatory reliability, and full operational sovereignty.

Chapter 2 — Business Problem

2. Business Problem

2.1 Context

The National Office for Food Safety (ONSSA) is responsible for publishing and maintaining a large collection of official regulatory documents governing food safety, sanitary regulations, veterinary regulations, phytosanitary measures, import and export procedures, administrative processes, technical guidelines, and legal obligations applicable to various stakeholders.

This documentary repository represents the official source of regulatory knowledge and includes a wide variety of document types, such as:

- Laws;
- Implementing decrees;
- Ministerial orders;
- Official Bulletins;
- Good practice guides;

- Administrative forms;
- Technical documentation;
- Regulatory web pages;
- Scanned documents;
- Multilingual documents (French and Arabic).

The repository continues to evolve as new legislation is published and existing regulations are updated or consolidated.

Although these documents are publicly accessible through the current ONSSA platform, users frequently face significant challenges when attempting to locate precise regulatory information efficiently.

2.2 Current Challenges

The current information access process is primarily document-oriented rather than knowledge-oriented.

Users typically need to:

- identify the relevant section of the ONSSA website;
- manually navigate multiple pages;
- open numerous PDF documents;
- search within lengthy legal texts;
- compare several regulatory documents;
- interpret relationships between laws, decrees, ministerial orders, and implementation guides.

This manual workflow is both time-consuming and error-prone, particularly when questions require combining information distributed across multiple official documents.

The complexity increases further because regulatory information is heterogeneous. Documents differ in format, structure, writing style, publication date, legal hierarchy, and language. Some documents are digitally structured, while others are scanned and require optical character recognition before becoming searchable.

As the documentary repository expands, traditional keyword-based search mechanisms become progressively less effective for answering complex regulatory questions.

2.3 Business Needs

The project addresses several strategic business needs.

Reliable access to official knowledge

Users require immediate access to authoritative regulatory information without manually reviewing dozens of documents.

Responses must always originate from official ONSSA documentation rather than from a model's internal knowledge.

Reduction of information retrieval time

Searching through hundreds of legal documents can require considerable effort.

An intelligent assistant should significantly reduce the time required to locate relevant regulatory information while preserving document authority.

Regulatory traceability

In a regulatory environment, answers alone are insufficient.

Every response must clearly indicate:

- the official document(s) consulted;
- the legal source;
- the exact passages supporting the answer.

This traceability enables users to verify every generated response against the original regulatory text.

Knowledge consistency

Different users may interpret regulations differently when consulting documents manually.

The project aims to provide standardized, consistent, and reproducible responses based exclusively on official documentation.

Confidentiality and Digital Sovereignty

Regulatory documents constitute institutional knowledge that must remain under ONSSA's control.

Consequently, the solution must operate entirely within ONSSA's infrastructure without transmitting documents or user queries to external cloud services or proprietary AI providers.

2.4 Problem Statement

ONSSA possesses a rich and authoritative regulatory knowledge base but lacks an intelligent mechanism capable of transforming this documentary repository into a searchable, explainable, and trustworthy knowledge service.

Traditional document search methods require users to manually identify, read, interpret, and correlate multiple legal texts before obtaining an answer. This process limits productivity, increases consultation time, and creates a risk of inconsistent interpretation.

At the same time, deploying a generic Large Language Model without access to official documentation would be unacceptable because such models may generate unsupported or hallucinated information, which is incompatible with regulatory decision-making.

The fundamental challenge is therefore not simply to build an AI assistant, but to design a system capable of combining the reasoning capabilities of a specialized Large Language Model with the authority of ONSSA's official documentary repository while guaranteeing traceability, transparency, confidentiality, and institutional control over all data and processing.

2.5 Proposed Solution Overview

To address these challenges, this project proposes the development of a specialized Artificial Intelligence service integrated into the existing ONSSA platform.

The solution will transform heterogeneous regulatory documents into a structured knowledge base through a centralized Knowledge Ingestion Pipeline. This knowledge base will subsequently support both model specialization and document retrieval.

Rather than relying solely on model memorization, the assistant will dynamically retrieve relevant regulatory information from the official Knowledge Corpus before generating each response. The specialized model will then synthesize the retrieved information into clear, structured answers while systematically providing references to the official documents used.

This architecture enables the system to deliver reliable, explainable, and evidence-based regulatory assistance while preserving the authority of official ONSSA documentation and ensuring that the AI service remains fully deployable within ONSSA's local infrastructure.

Chapter 3 — Vision

3. Vision

3.1 Product Vision

The vision of this project is to establish a trusted Artificial Intelligence service that transforms ONSSA's regulatory documentation into an intelligent, searchable, and explainable institutional knowledge system while preserving the authority of official regulatory documents.

Rather than replacing the existing ONSSA platform, the proposed solution enhances its capabilities by providing an AI-powered regulatory assistant that enables users to interact with official documentation through natural language queries.

The assistant is designed to provide reliable answers grounded exclusively in official ONSSA documentation. Every generated response must remain verifiable through explicit references to the regulatory sources used during answer generation.

The long-term objective is to progressively transform the entire ONSSA documentary repository into an intelligent knowledge service while maintaining full compliance with institutional security, confidentiality, and digital sovereignty requirements.

3.2 Strategic Vision

The project is intended to become the intelligent regulatory layer of the existing ONSSA information system.

Instead of acting as an isolated application, the AI service operates as a reusable backend component that can be integrated into multiple ONSSA digital services through standardized REST APIs.

This architectural approach ensures that future ONSSA applications can leverage the same centralized regulatory intelligence without duplicating business logic or maintaining separate knowledge repositories.

The solution therefore represents an institutional capability rather than a standalone software application.

3.3 Guiding Principles

The design of the system follows several fundamental principles that govern every architectural decision throughout this project.

Official documentation remains the single source of truth

The system does not generate regulatory knowledge independently.

All factual information must originate from official ONSSA documentation processed through the Knowledge Ingestion Pipeline.

The Large Language Model is responsible for interpreting and presenting retrieved information rather than replacing the documentary authority of legal texts.

Explainability by design

Every response produced by the system must be traceable to its supporting regulatory documents.

Users should always be able to identify:

- the official documents consulted;
- the regulatory sources supporting the answer;
- the evidence used during response generation.

Explainability is considered a mandatory system characteristic rather than an optional feature.

Human-centric knowledge access

The objective is not to simplify regulations by removing legal precision.

Instead, the system aims to make complex regulatory information easier to discover, understand, and navigate while preserving the original meaning of official documents.

The assistant should reduce the effort required to access knowledge without altering regulatory content.

Digital sovereignty

All processing stages—including document ingestion, knowledge management, model specialization, inference, retrieval, and deployment—must remain entirely under ONSSA's control.

The solution is designed for fully local deployment without dependence on external cloud-based Artificial Intelligence services.

Institutional knowledge never leaves the ONSSA infrastructure.

Scalable architecture

The project is intentionally designed as a scalable architecture rather than a single-use implementation.

The same architecture should support future extensions covering additional ONSSA regulatory domains without requiring significant redesign.

New document collections should be incorporated by extending the knowledge ingestion process rather than modifying the core system architecture.

3.4 Vision for Knowledge Management

Beyond developing an AI assistant, the project aims to establish a sustainable institutional knowledge management framework.

The Knowledge Ingestion Pipeline serves as the foundation for continuously transforming heterogeneous regulatory documents into structured, machine-readable knowledge assets.

These knowledge assets become reusable across multiple AI capabilities, including:

- document retrieval;
- semantic search;
- regulatory question answering;
- future AI services;
- institutional knowledge management initiatives.

This approach ensures that document processing is performed once while enabling multiple downstream applications to benefit from the resulting Knowledge Corpus.

3.5 Success Vision

At the completion of the project, ONSSA will possess a production-ready Artificial Intelligence service capable of:

- understanding regulatory questions expressed in natural language;
- retrieving relevant information from official ONSSA documentation;
- generating grounded and explainable responses;
- systematically citing the regulatory sources used;
- operating entirely within ONSSA's local infrastructure;
- integrating seamlessly into the existing ONSSA platform through a REST API.

The solution will establish a reusable technological foundation that enables ONSSA to progressively extend intelligent access to its complete regulatory repository while preserving document authority, transparency, security, and long-term maintainability.

Chapter 4 — Objectives

4. Objectives

4.1 Project Objective

The primary objective of this project is to develop a production-ready Artificial Intelligence service capable of providing accurate, explainable, and evidence-based answers to regulatory questions using exclusively official ONSSA documentation.

The solution is designed to integrate seamlessly into the existing ONSSA platform through a REST API while operating entirely within ONSSA's local infrastructure. The system must enable users to query regulatory information using natural language and receive responses that are grounded in official documents and accompanied by precise source references.

The project does not aim to replace existing regulatory documentation or established consultation procedures. Instead, it enhances access to institutional knowledge by combining modern Artificial Intelligence techniques with the authority of ONSSA's official documentary repository.

4.2 Business Objectives

The project pursues several strategic business objectives that directly address the operational challenges identified in the previous chapter.

Improve Access to Regulatory Knowledge

Provide users with rapid and intuitive access to official regulatory information without requiring manual exploration of numerous documents.

The system should significantly reduce the effort needed to locate relevant regulations while preserving the legal authority of the original texts.

Increase Operational Efficiency

Reduce the time required to answer regulatory questions by automating document retrieval and knowledge synthesis.

By minimizing repetitive document searches, the solution allows personnel to focus on regulatory analysis and decision-making rather than information retrieval.

Improve Knowledge Consistency

Ensure that similar regulatory questions consistently produce coherent answers based on the same official sources.

The assistant should reduce variability caused by manual interpretation and document navigation.

Strengthen Regulatory Traceability

Guarantee that every generated answer includes explicit references to the supporting regulatory documents.

Traceability enhances transparency, facilitates verification, and reinforces user confidence in AI-assisted responses.

Preserve Institutional Sovereignty

Ensure that all regulatory documents, knowledge assets, models, and user interactions remain under ONSSA's direct control.

The complete solution must be deployable within ONSSA's infrastructure without relying on external Artificial Intelligence services.

4.3 Technical Objectives

To achieve the business objectives, the project defines the following technical objectives.

Build a Centralized Knowledge Ingestion Pipeline

Develop a robust pipeline capable of discovering, collecting, processing, cleaning, structuring, and organizing heterogeneous ONSSA documents into a unified Knowledge Corpus.

The pipeline constitutes the foundation of the entire system and serves as the single source of structured knowledge for downstream components.

Produce Reusable Knowledge Assets

Generate three complementary outputs from the same documentary source:

- a structured Knowledge Corpus;
- an Instruction Dataset for model specialization;

- vector embeddings supporting semantic retrieval.

This design ensures that all downstream AI components rely on a common and consistent knowledge foundation.

Specialize the Language Model

Adapt a locally deployed open-source Large Language Model using parameter-efficient fine-tuning techniques.

The objective of model specialization is not to memorize regulations but to improve:

- regulatory language understanding;
- ONSSA-specific terminology;
- response organization;
- citation formatting;
- instruction-following behavior.

Factual regulatory information remains external to the model and is retrieved dynamically during inference.

Implement Reliable Knowledge Retrieval

Provide an efficient retrieval mechanism capable of identifying the most relevant document fragments for each user query.

Retrieved information must originate exclusively from the structured Knowledge Corpus generated by the ingestion pipeline.

Generate Explainable Responses

Produce responses that combine the reasoning capabilities of the specialized model with the factual accuracy of retrieved regulatory documents.

Every answer must remain fully grounded in official documentation and include supporting references.

Provide a Production-Ready Service

Expose the complete AI capability through a secure REST API suitable for integration into the existing ONSSA platform.

The service should support future extensions without requiring major architectural changes.

4.4 Incremental Development Objectives

The project follows an incremental implementation strategy in order to minimize technical risk and validate the architecture before extending it to the complete ONSSA repository.

Objective 1 — Validate the Complete Pipeline

The first milestone consists of validating the entire end-to-end architecture using a limited but representative subset of regulatory documents corresponding to:

Regulation → Transversal Regulation → Food Safety

This Vertical Slice demonstrates the feasibility of:

- document acquisition;
- document processing;
- knowledge corpus construction;
- dataset generation;
- embedding creation;
- model specialization;
- retrieval;
- answer generation;
- evaluation;
- deployment;
- integration.

Completion of this milestone confirms that the proposed architecture functions correctly under production-like conditions.

Objective 2 — Extend the Knowledge Coverage

Once the Vertical Slice has been successfully validated, the same architecture will be progressively extended to additional ONSSA regulatory domains.

Expansion consists primarily of processing new document collections through the existing Knowledge Ingestion Pipeline without redesigning the system architecture.

This approach enables controlled growth while preserving consistency across the entire knowledge base.

4.5 Success Criteria

The project will be considered successful when the following objectives have been achieved:

- Official ONSSA documents are transformed into a structured Knowledge Corpus through the complete Knowledge Ingestion Pipeline.
- The generated Knowledge Corpus successfully supports both model specialization and semantic retrieval.
- The specialized language model accurately follows ONSSA-specific response conventions and citation requirements.
- The retrieval mechanism consistently identifies relevant regulatory information from the official knowledge base.
- Generated responses remain grounded in official documentation and include verifiable references.
- The complete AI service operates entirely within ONSSA's local infrastructure.
- The service is accessible through a REST API and can be integrated into the existing ONSSA platform without modifying its overall architecture.
- The validated Vertical Slice demonstrates that the proposed architecture can be scaled incrementally to the remainder of the ONSSA documentary repository.

Chapter 5 — Scope

5. Scope

5.1 Project Scope

This project focuses on the design, development, evaluation, and deployment of a specialized Artificial Intelligence service dedicated to regulatory knowledge retrieval within the existing ONSSA information system.

The solution is not intended to replace the current ONSSA platform. Instead, it provides an independent AI backend service that integrates seamlessly with the existing platform through a REST API.

The project encompasses the complete lifecycle required to transform heterogeneous regulatory documents into a reliable, searchable, and explainable knowledge service.

The implementation covers the complete end-to-end architecture, from document acquisition to response generation and deployment.

5.2 Functional Scope

The project includes the development of the following functional components.

Knowledge Acquisition

The system shall automatically acquire official ONSSA documentation from the designated documentary sources.

Supported document types include:

- HTML web pages;
- PDF documents;
- Microsoft Word documents;
- Official Bulletins;
- Good Practice Guides;
- Laws;
- Implementing Decrees;
- Ministerial Orders;
- Administrative Forms;
- Scanned documents;
- Images containing regulatory information.

Document acquisition is performed through the Knowledge Ingestion Pipeline.

Knowledge Processing

The project includes the complete document processing workflow required to transform heterogeneous documents into structured knowledge assets.

Processing activities include:

- document discovery;
- document downloading;
- HTML parsing;
- PDF parsing;
- OCR processing when required;
- document cleaning;
- duplicate detection;
- metadata extraction;
- document classification;
- legal structure extraction;
- intelligent chunk generation.

The resulting structured information forms the Knowledge Corpus that serves as the single source of knowledge throughout the system.

Knowledge Asset Generation

The Knowledge Corpus generated by the ingestion pipeline is used to produce two complementary knowledge assets.

The first is an Instruction Dataset dedicated to parameter-efficient model specialization.

The second consists of semantic vector embeddings supporting document retrieval during inference.

Both assets originate from the same authoritative Knowledge Corpus, ensuring consistency across the entire architecture.

Model Specialization

The project includes the specialization of a locally deployed open-source Large Language Model using parameter-efficient fine-tuning.

Model specialization focuses on:

- ONSSA regulatory terminology;
- response organization;
- citation formatting;
- instruction following;
- professional communication style.

The model is not trained to memorize the regulatory corpus.

Domain knowledge remains external and is retrieved dynamically through the retrieval system.

Retrieval-Augmented Generation

The project includes the implementation of a Retrieval-Augmented Generation (RAG) architecture capable of:

- understanding user questions;
- retrieving relevant regulatory passages;
- combining retrieved evidence with the specialized language model;
- generating grounded responses;
- providing explicit citations to supporting documents.

The retrieval process operates exclusively on the structured Knowledge Corpus generated by the ingestion pipeline.

Evaluation

The project includes the design and implementation of an evaluation framework capable of assessing:

- retrieval quality;
- response accuracy;
- citation correctness;
- hallucination resistance;
- instruction-following capability;

- overall system performance.

Evaluation is considered an integral component of the production workflow rather than a post-development activity.

Deployment

The project includes a complete local deployment architecture based on containerized services.

Deployment covers:

- model serving;
- vector database;
- REST API;
- orchestration;
- configuration;
- integration with the existing ONSSA platform.

No external AI service is required during production.

5.3 Vertical Slice Scope

To reduce implementation risk and validate the architecture incrementally, the first development phase intentionally focuses on a limited yet representative subset of ONSSA documentation.

The initial Vertical Slice includes only the following regulatory domain:

Regulation → Transversal Regulation → Food Safety

All components of the proposed architecture—including document acquisition, processing, corpus generation, model specialization, semantic retrieval, response generation, evaluation, deployment, and platform integration—must be fully operational within this restricted scope before any expansion is considered.

This approach validates the complete end-to-end workflow under realistic production conditions while limiting project complexity.

5.4 Deliverables

The project delivers the following production-ready components:

Knowledge Ingestion Pipeline

A complete pipeline capable of automatically transforming heterogeneous ONSSA documents into structured knowledge assets.

Knowledge Corpus

A normalized and structured JSON-based knowledge repository serving as the authoritative source for downstream AI components.

Instruction Dataset

A high-quality instruction dataset generated from the Knowledge Corpus to support parameter-efficient model specialization.

Vector Knowledge Base

A semantic vector index built from the Knowledge Corpus to enable efficient document retrieval during inference.

Specialized Language Model

A locally deployed Large Language Model specialized in ONSSA regulatory communication through parameter-efficient fine-tuning.

Retrieval-Augmented Generation Engine

A retrieval system capable of grounding model responses in official regulatory documentation.

Evaluation Framework

A comprehensive evaluation framework measuring both retrieval effectiveness and generation quality.

AI Service API

A production-ready REST API exposing the AI capabilities for consumption by the existing ONSSA platform.

Deployment Environment

A fully containerized deployment architecture using local infrastructure, enabling secure and reproducible installation within ONSSA.

5.5 Scope Boundaries

This project delivers the complete technical foundation required for an institutional regulatory AI service.

The architecture is intentionally designed to support future expansion without requiring major redesign.

However, only the Vertical Slice defined in this chapter is implemented during the initial development phase.

Additional regulatory domains, document collections, and institutional knowledge sources will be incorporated in subsequent phases by extending the existing Knowledge Ingestion Pipeline and regenerating the associated knowledge assets while preserving the overall system architecture.

Chapter 6 — Out of Scope

6. Out of Scope

To ensure a controlled implementation and validate the proposed architecture under realistic conditions, several capabilities are intentionally excluded from the scope of this project.

These exclusions do not represent technical limitations of the proposed architecture. Instead, they reflect deliberate project boundaries established to prioritize the successful validation of the complete end-to-end workflow before progressively extending the solution.

6.1 Full ONSSA Repository Coverage

The project does not aim to process the complete ONSSA documentary repository during the initial implementation.

Although the proposed architecture is designed to scale to all ONSSA regulatory domains, the first implementation is intentionally restricted to the following Vertical Slice:

Regulation → Transversal Regulation → Food Safety

Only after validating the complete architecture on this representative subset will additional document collections be incorporated using the same Knowledge Ingestion Pipeline.

6.2 Development of a New Web Platform

The project does not include the design or development of a new user interface, web portal, or standalone application.

The existing ONSSA platform remains the primary user interface.

The deliverable is an independent Artificial Intelligence service exposing a REST API that can be consumed by the current platform without modifying its overall architecture.

6.3 Replacement of Existing Information Systems

The proposed solution does not replace existing document management systems, content management systems, databases, or regulatory repositories.

Official documents remain stored and managed according to ONSSA's existing governance processes.

The AI service complements these systems by providing intelligent knowledge access rather than replacing them.

6.4 Legal Interpretation and Decision Making

The system is not intended to provide legal advice or make regulatory decisions.

Its responsibility is limited to retrieving relevant information from official documentation and generating grounded responses supported by explicit citations.

Final interpretation, regulatory validation, and decision-making remain the responsibility of qualified ONSSA personnel.

6.5 Automatic Modification of Official Documents

The project does not modify, rewrite, summarize for publication, or replace official regulatory documents.

The documentary repository remains the authoritative legal reference.

The AI service only processes documents internally to construct structured knowledge assets while preserving the integrity of the original sources.

6.6 Continuous Legislative Monitoring

Automatic monitoring of new legislation, real-time regulatory updates, or continuous synchronization with external legal publication systems is outside the scope of the initial implementation.

The first version assumes controlled execution of the Knowledge Ingestion Pipeline whenever new official documents are made available.

Future versions may introduce scheduled or event-driven ingestion workflows.

6.7 Multimodal Question Answering

Although the ingestion pipeline processes images and scanned documents through OCR when necessary, the first version is not intended to answer questions directly about visual content such as:

- diagrams;
- handwritten annotations;
- graphical illustrations;
- maps;
- complex figures.

The objective is to extract textual regulatory information contained within those documents.

6.8 Memorization of the Regulatory Corpus

The project does not attempt to train the language model to memorize ONSSA regulations.

This is a deliberate architectural decision.

Regulatory knowledge remains external to the model and is retrieved dynamically from the Knowledge Corpus through the Retrieval-Augmented Generation architecture.

The Fine-Tuning process is limited to improving:

- domain terminology;
 - response style;
 - instruction-following capability;
 - citation formatting;
 - regulatory communication behavior.
-

6.9 Cloud-Based Artificial Intelligence Services

The project excludes the use of external cloud-hosted Large Language Models or proprietary AI APIs for production inference.

Production deployment relies exclusively on locally hosted open-source models executed within ONSSA's infrastructure.

This decision preserves confidentiality, institutional sovereignty, and full control over regulatory data.

6.10 General-Purpose Conversational Assistant

The system is not designed to function as a general-purpose chatbot.

Its capabilities are intentionally restricted to ONSSA regulatory knowledge contained within the processed documentary repository.

Questions outside the available knowledge base fall outside the intended operational scope.

When sufficient evidence cannot be retrieved from official documentation, the assistant must explicitly indicate that it cannot provide a reliable answer rather than generating unsupported information.

6.11 Autonomous Knowledge Creation

The assistant is not authorized to generate new regulatory knowledge, infer unpublished rules, or speculate about legal interpretations.

Every factual statement included in a response must be supported by information retrieved from official ONSSA documentation.

The system therefore prioritizes factual reliability and traceability over conversational completeness.

6.12 First-Version Boundary

The first production release focuses exclusively on validating the complete technical architecture through a representative regulatory subset.

The following objectives are intentionally deferred to subsequent project phases:

- extension to additional ONSSA regulatory domains;
- ingestion of the complete documentary repository;
- automated document update scheduling;
- broader multilingual optimization;
- advanced analytics and reporting capabilities;
- additional AI services built upon the same Knowledge Corpus.

This incremental strategy minimizes implementation risk while ensuring that each architectural layer is fully validated before the solution is expanded institution-wide.

Chapter 7 — Functional Requirements

7. Functional Requirements

7.1 Overview

The proposed solution shall provide an end-to-end Artificial Intelligence service capable of transforming official ONSSA regulatory documents into a searchable institutional knowledge system and delivering evidence-based responses through the existing ONSSA platform.

The functional architecture follows a sequential workflow in which each component produces artifacts consumed by the subsequent stages of the system.

The functional requirements defined in this chapter describe **what** the system must accomplish. The technical implementation of each capability is specified in the architecture chapters that follow.

FR-1 Document Discovery

Objective

The system shall automatically identify official regulatory documents belonging to the defined ONSSA documentary scope.

Description

The discovery process represents the entry point of the Knowledge Ingestion Pipeline.

It is responsible for identifying all documents that belong to the selected regulatory domain while avoiding unrelated or unsupported resources.

For the initial Vertical Slice, discovery is restricted to:

Regulation → Transversal Regulation → Food Safety

The discovery mechanism shall support future extension to additional ONSSA repositories without redesigning the overall architecture.

Expected Output

A validated list of discoverable regulatory resources ready for acquisition.

FR-2 Document Acquisition

Objective

The system shall acquire all discovered regulatory documents.

Description

The acquisition component downloads or retrieves the documents identified during the discovery phase while preserving their integrity.

Supported document categories include:

- HTML pages
- PDF documents
- Word documents
- Official Bulletins
- Laws
- Decrees
- Ministerial Orders
- Good Practice Guides
- Administrative Forms
- Scanned Documents

Document acquisition must preserve sufficient metadata to enable downstream processing.

Expected Output

A local collection of raw regulatory documents ready for processing.

FR-3 Document Processing

Objective

The system shall transform heterogeneous raw documents into structured textual content.

Description

The processing workflow shall support multiple document formats while extracting textual information suitable for downstream knowledge construction.

The system shall be capable of:

- parsing HTML documents;
- parsing PDF files;
- extracting textual content;
- executing OCR when textual layers are unavailable;
- preserving document structure whenever possible.

The objective is to normalize heterogeneous document formats into a common representation.

Expected Output

Normalized textual representations of all processed documents.

FR-4 Knowledge Preparation

Objective

The system shall prepare high-quality regulatory content for knowledge extraction.

Description

The preparation stage shall improve the quality and consistency of extracted information before knowledge structuring.

The workflow shall include:

- document cleaning;
- duplicate detection;
- removal of irrelevant content;
- metadata extraction;
- document categorization;
- preservation of legal hierarchy.

Knowledge preparation shall ensure that downstream AI components operate on reliable and standardized information.

Expected Output

Validated regulatory content enriched with document metadata.

FR-5 Knowledge Structuring

Objective

The system shall transform processed regulatory documents into structured knowledge assets.

Description

The system shall identify and preserve the logical organization of regulatory documents, including legal hierarchy and document organization.

The structured representation shall support:

- article-level organization;
- section hierarchy;
- legal references;
- document relationships;
- intelligent semantic segmentation.

This stage concludes with the construction of the centralized Knowledge Corpus.

Expected Output

A structured Knowledge Corpus representing the official regulatory knowledge base.

FR-6 Knowledge Asset Generation

Objective

The system shall derive reusable AI knowledge assets from the Knowledge Corpus.

Description

The Knowledge Corpus constitutes the single authoritative knowledge source for downstream AI components.

From this corpus, the system shall generate:

- an Instruction Dataset for model specialization;
- semantic embeddings for retrieval.

Both assets shall remain synchronized because they originate from the same structured corpus.

Expected Output

- Knowledge Corpus
 - Instruction Dataset
 - Embedding Dataset
-

FR-7 LLM Specialization through QLoRA Fine-Tuning

Objective

The system shall specialize a locally deployed Large Language Model for ONSSA regulatory communication.

Description

Model specialization shall improve:

- regulatory terminology;
- institutional vocabulary;
- response formatting;
- citation behavior;
- instruction following.

The model shall **not** memorize the regulatory corpus.

Its specialization is limited to behavioral adaptation.

Regulatory knowledge remains external to the model.

Expected Output

A specialized ONSSA language model.

FR-8 Semantic Retrieval

Objective

The system shall retrieve the most relevant regulatory information for each user query.

Description

The retrieval component searches the vector database built from the Knowledge Corpus to identify document fragments that are semantically relevant to the user's request.

Retrieved information becomes the factual context supplied to the language model.

The retrieval component shall operate exclusively on official ONSSA knowledge assets.

Expected Output

A ranked collection of relevant regulatory passages.

FR-9 Grounded Response Generation

Objective

The system shall generate responses grounded in retrieved regulatory evidence.

Description

The specialized language model combines:

- user query;
- retrieved document passages;
- system instructions.

The generated response shall remain faithful to retrieved evidence.

When insufficient evidence is available, the system shall explicitly state that it cannot answer reliably.

Expected Output

A grounded regulatory response.

FR-10 Citation Generation

Objective

The system shall provide transparent traceability for every generated response.

Description

Every answer shall include references to the regulatory documents used during generation.

Whenever possible, citations shall identify:

- source document;
- document type;
- article or section;
- legal reference.

Citation generation is mandatory for every supported response.

Expected Output

Response accompanied by verifiable regulatory citations.

FR-11 System Evaluation

Objective

The system shall evaluate its retrieval and generation performance.

Description

Evaluation shall assess the complete AI pipeline, including:

- retrieval quality;
- response quality;
- citation accuracy;
- hallucination resistance;
- instruction compliance;
- overall reliability.

Evaluation results shall support iterative improvement of both the Knowledge Corpus and the AI models.

Expected Output

Comprehensive evaluation reports and performance metrics.

FR-12 AI Service Exposure

Objective

The system shall expose its capabilities through a production-ready REST API.

Description

The API serves as the integration point between the AI service and the existing ONSSA platform.

The service shall receive user queries, orchestrate the complete AI workflow, and return grounded responses with supporting citations.

The API layer shall remain independent from the presentation layer, allowing multiple ONSSA applications to consume the same AI capabilities.

Expected Output

A secure REST API exposing the complete regulatory assistant workflow.

Chapter 8 — Non-Functional Requirements

8. Non-Functional Requirements

8.1 Overview

Beyond functional capabilities, the proposed AI service must satisfy a set of non-functional requirements that ensure reliability, security, maintainability, scalability, and operational readiness within ONSSA's information system.

These requirements define the quality attributes of the solution and establish the constraints that guide all architectural and implementation decisions.

As the system is intended for institutional deployment and operates on official regulatory documents, non-functional requirements are considered first-class engineering objectives rather than secondary considerations.

8.2 Performance

NFR-1 Response Time

The system shall provide interactive response times suitable for everyday use through the existing ONSSA platform.

Performance optimization shall consider the complete inference workflow, including:

- semantic retrieval;
- prompt construction;
- LLM inference;
- citation generation;
- response formatting.

Performance shall remain consistent regardless of document origin or format.

NFR-2 Knowledge Retrieval Efficiency

The retrieval component shall efficiently identify relevant regulatory passages from the vector database while maintaining high retrieval accuracy.

The retrieval process shall scale with the growth of the Knowledge Corpus without requiring architectural modifications.

NFR-3 Pipeline Throughput

The Knowledge Ingestion Pipeline shall efficiently process large collections of heterogeneous documents.

Processing performance shall support incremental corpus expansion as additional ONSSA regulatory domains are incorporated.

Pipeline execution is expected to operate as a background process independent of user interactions.

8.3 Reliability

NFR-4 Response Reliability

Responses shall be generated exclusively from retrieved official documentation.

The system shall prioritize factual correctness over conversational completeness.

Whenever sufficient supporting evidence cannot be retrieved, the assistant shall explicitly communicate this limitation instead of generating speculative information.

NFR-5 Citation Reliability

Every supported response shall include references to the regulatory documents used during generation.

Citation generation is mandatory and shall not be treated as an optional enhancement.

NFR-6 Fault Tolerance

Failures affecting individual documents during the ingestion process shall not interrupt processing of the remaining document collection.

The pipeline shall continue processing unaffected resources while reporting failed operations for subsequent review.

8.4 Security

NFR-7 Local Deployment

The entire solution shall operate within ONSSA's infrastructure.

No production data, regulatory documents, embeddings, prompts, or user queries shall be transmitted to external Artificial Intelligence services.

NFR-8 Data Confidentiality

All processed documents, intermediate artifacts, generated datasets, vector indexes, and specialized models shall remain under ONSSA's administrative control.

Access to internal system components shall follow ONSSA's security policies.

NFR-9 API Security

The REST API shall support secure communication mechanisms appropriate for integration with institutional applications.

Authentication, authorization, and transport security shall be configurable according to ONSSA's infrastructure standards.

8.5 Scalability

NFR-10 Corpus Scalability

The architecture shall support progressive expansion from the initial Vertical Slice to the complete ONSSA documentary repository.

Extending the corpus shall primarily require executing the existing Knowledge Ingestion Pipeline on additional document collections without redesigning downstream components.

NFR-11 Modular Expansion

Each architectural component shall evolve independently whenever possible.

Future improvements to document processing, retrieval algorithms, embedding models, or language models shall not require redesign of unrelated system components.

8.6 Maintainability

NFR-12 Modular Architecture

The solution shall follow a modular architecture in which each major component has clearly defined responsibilities and interfaces.

Core architectural modules include:

- Knowledge Ingestion Pipeline;
- Knowledge Corpus;
- Instruction Dataset generation;
- Embedding generation;
- Model specialization;
- Retrieval engine;
- Response generation;
- Evaluation framework;
- API service.

This modular organization simplifies maintenance and future evolution.

NFR-13 Reproducibility

The complete document processing workflow shall produce reproducible knowledge assets when executed on the same documentary sources using identical configurations.

This requirement facilitates debugging, auditing, and future corpus regeneration.

NFR-14 Configuration Management

System behavior shall be configurable through external configuration rather than source code modifications.

Examples include:

- document sources;
 - processing parameters;
 - chunking strategy;
 - embedding models;
 - retrieval parameters;
 - inference settings.
-

8.7 Data Quality

NFR-15 Knowledge Integrity

The Knowledge Corpus shall accurately represent the information contained within official ONSSA documentation.

Document processing shall preserve legal structure and contextual relationships whenever possible.

NFR-16 Metadata Consistency

Every processed document shall contain standardized metadata required for retrieval, citation generation, filtering, and future corpus management.

Metadata consistency is essential for downstream AI components.

NFR-17 Traceability

Every generated knowledge asset shall remain traceable to its originating official document.

This traceability shall be preserved throughout the complete processing pipeline, from raw document acquisition to final response generation.

8.8 Explainability

NFR-18 Transparent Responses

Generated responses shall clearly distinguish factual regulatory information from explanatory language produced by the language model.

Users shall always be able to identify the supporting documentary evidence.

NFR-19 Evidence-Based Generation

The assistant shall generate responses exclusively from retrieved regulatory evidence.

The architecture shall minimize unsupported generation by grounding inference on the Knowledge Corpus.

8.9 Deployment

NFR-20 Containerized Deployment

All production services shall support containerized deployment.

Containerization simplifies installation, version management, portability, and future maintenance.

NFR-21 Local AI Infrastructure

Inference, retrieval, vector indexing, document processing, and API services shall execute entirely within ONSSA's local infrastructure.

The production environment shall not depend on Internet connectivity for AI inference.

NFR-22 Integration Readiness

The AI service shall expose standardized REST interfaces that allow seamless integration with the existing ONSSA platform.

The service shall remain independent of any specific presentation layer.

8.10 Observability

NFR-23 Logging

The system shall generate structured logs for major processing stages, including:

- document ingestion;
- pipeline execution;
- retrieval operations;
- inference requests;
- API interactions;
- system errors.

Logs shall facilitate troubleshooting and operational monitoring.

NFR-24 Monitoring

The production environment shall support monitoring of critical operational indicators, including:

- service availability;
- processing status;
- inference latency;
- retrieval latency;
- API performance;
- resource utilization.

Monitoring enables proactive maintenance and rapid incident detection.

8.11 Extensibility

NFR-25 Future Expansion

The architecture shall support future extension to:

- additional ONSSA regulatory domains;
- new document formats;
- improved embedding models;
- newer language models;
- enhanced retrieval strategies;
- additional institutional AI services.

These extensions shall leverage the existing Knowledge Ingestion Pipeline and Knowledge Corpus without requiring fundamental architectural redesign.

8.12 Summary

The non-functional requirements establish the engineering principles that govern the entire solution.

Together, they ensure that the proposed AI service is not only functionally correct but also secure, reliable, maintainable, scalable, explainable, and suitable for long-term deployment within ONSSA's information system.

These quality attributes directly influence the architectural decisions presented in the following chapters, beginning with the overall system architecture.

Chapter 9 — System Architecture

9. System Architecture

9.1 Architectural Overview

The proposed solution is designed as a modular Artificial Intelligence service that integrates with the existing ONSSA platform while remaining logically independent from the presentation layer.

The architecture follows a pipeline-oriented design in which regulatory documents are progressively transformed into structured knowledge assets that support both model specialization and Retrieval-Augmented Generation (RAG).

Rather than relying on the language model to memorize regulatory knowledge, the system separates **knowledge construction**, **knowledge retrieval**, and **language generation** into independent architectural layers.

This separation improves maintainability, scalability, explainability, and long-term evolution of the system.

The overall architecture is organized into the following major layers:

1. Document Sources
2. Knowledge Ingestion Pipeline
3. Knowledge Corpus
4. Knowledge Asset Generation
5. Model Specialization
6. Knowledge Retrieval
7. Response Generation
8. Evaluation
9. Deployment Layer
10. Integration Layer

Each layer has a single responsibility and produces artifacts consumed by the next stage of the architecture.

9.2 Architectural Principles

The architecture has been designed according to several fundamental engineering principles.

Separation of Responsibilities

Each major component performs a single well-defined responsibility.

For example:

- the Knowledge Ingestion Pipeline constructs knowledge;
- the Knowledge Corpus stores structured knowledge;
- the Instruction Dataset specializes the language model;
- the Embedding Dataset supports semantic retrieval;
- the Vector Database retrieves knowledge;
- the LLM generates responses;
- the Evaluation Framework measures system quality.

This separation minimizes coupling between components and facilitates future maintenance.

Single Source of Truth

The Knowledge Corpus is the authoritative representation of ONSSA regulatory knowledge.

Every downstream AI component derives its information from this centralized corpus.

Neither the Fine-Tuned model nor the Vector Database independently defines regulatory knowledge.

Instead, both are generated from the same Knowledge Corpus, ensuring consistency across the architecture.

Progressive Knowledge Transformation

Knowledge is progressively refined through successive processing stages.

Rather than directly feeding raw documents into the language model, the system incrementally transforms them into structured knowledge assets optimized for different AI tasks.

This staged transformation improves quality, traceability, and maintainability.

Explainability by Design

Every generated answer must remain traceable to official regulatory documents.

Explainability is therefore embedded throughout the architecture rather than added as a post-processing feature.

Document metadata, legal hierarchy, and source references are preserved from the earliest processing stages to final response generation.

Local-First Deployment

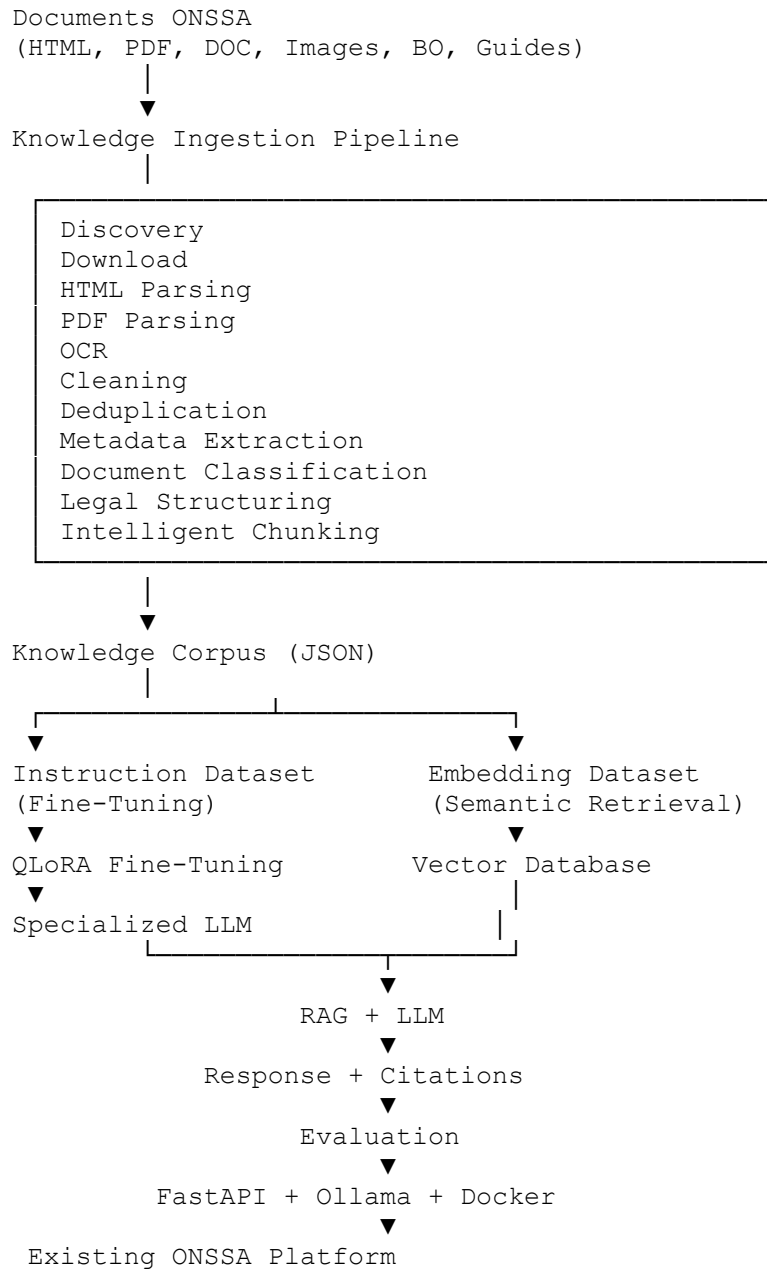
All architectural components are designed for execution within ONSSA's local infrastructure.

No architectural dependency exists on external Artificial Intelligence services.

This principle guarantees confidentiality, institutional sovereignty, and operational independence.

9.3 High-Level Architecture

The complete system follows the architecture illustrated below.



This architecture constitutes the reference architecture for the remainder of this document.

Subsequent chapters specify each architectural layer without modifying this overall design.

9.4 Architectural Layers

Layer 1 — Document Sources

The first architectural layer consists of the official ONSSA documentary repository.

Supported sources include:

- HTML pages;
- PDF documents;
- Microsoft Word documents;
- Official Bulletins;
- Laws;
- Decrees;
- Ministerial Orders;
- Good Practice Guides;
- Administrative Forms;
- Images;
- Scanned documents.

These documents represent the only authoritative source of regulatory knowledge.

Layer 2 — Knowledge Ingestion Pipeline

The Knowledge Ingestion Pipeline is the core architectural component of the entire solution.

Its responsibility is to transform heterogeneous regulatory documents into structured knowledge suitable for Artificial Intelligence applications.

The pipeline performs the complete document engineering workflow:

- discovery;
- acquisition;
- parsing;
- OCR;
- cleaning;
- deduplication;
- metadata extraction;
- document classification;
- legal structuring;
- intelligent chunking.

The output of this layer is the centralized Knowledge Corpus.

No downstream AI component accesses raw documents directly.

Layer 3 — Knowledge Corpus

The Knowledge Corpus represents the canonical machine-readable representation of ONSSA regulatory knowledge.

It is implemented as a structured JSON repository that preserves:

- document hierarchy;
- legal organization;
- metadata;
- document relationships;
- chunk boundaries;
- source traceability.

The Knowledge Corpus is the single source of truth for all downstream processing.

Layer 4 — Knowledge Asset Generation

From the Knowledge Corpus, the architecture generates two complementary knowledge assets.

Instruction Dataset

The Instruction Dataset is created to specialize the language model through parameter-efficient fine-tuning.

Its purpose is to teach:

- regulatory communication;
- institutional vocabulary;
- response formatting;
- citation conventions;
- instruction-following behavior.

It is **not** intended to memorize regulatory content.

Embedding Dataset

The same Knowledge Corpus is converted into semantic vector representations.

These embeddings populate the Vector Database and support semantic document retrieval during inference.

Using a common source guarantees consistency between model specialization and retrieval.

Layer 5 — Model Specialization

The language model is specialized through QLoRA Fine-Tuning.

Specialization modifies the model's behavior rather than its factual knowledge.

After specialization, the model understands:

- ONSSA terminology;
- regulatory writing style;
- response structure;
- citation formatting;
- interaction guidelines.

The model continues to rely on external knowledge retrieval for regulatory facts.

Layer 6 — Retrieval-Augmented Generation

When a user submits a query, the Retrieval-Augmented Generation architecture performs the following sequence:

1. Interpret the user query.
2. Search the Vector Database.
3. Retrieve relevant document chunks.
4. Construct the retrieval context.
5. Generate a grounded response using the specialized language model.
6. Attach supporting citations.

This architecture minimizes hallucinations by grounding generation in official regulatory evidence.

Layer 7 — Evaluation Framework

Evaluation is integrated as a permanent architectural component rather than a final project activity.

The evaluation layer measures:

- retrieval effectiveness;
- generation quality;
- citation correctness;
- grounding quality;
- instruction compliance;
- hallucination rate;

- overall system performance.

Evaluation continuously validates both the Knowledge Corpus and the AI models.

Layer 8 — Deployment Layer

The production environment packages the AI service as containerized components.

The deployment architecture includes:

- Specialized LLM;
- Vector Database;
- FastAPI service;
- Ollama runtime;
- supporting services.

Containerization ensures portability, reproducibility, and simplified maintenance.

Layer 9 — Integration Layer

The AI service exposes standardized REST endpoints that integrate with the existing ONSSA platform.

The platform remains responsible for:

- authentication;
- user interface;
- session management;
- user workflows.

The AI service is responsible exclusively for regulatory intelligence.

This separation allows future ONSSA applications to reuse the same AI backend without duplicating business logic.

9.5 Architectural Workflow

The end-to-end workflow of the system can be summarized as follows:

1. Official ONSSA documents are collected.
2. The Knowledge Ingestion Pipeline processes the documents.
3. A structured Knowledge Corpus is generated.

4. The corpus produces both the Instruction Dataset and semantic embeddings.
5. The language model is specialized using QLoRA.
6. The embeddings populate the Vector Database.
7. User queries trigger semantic retrieval.
8. Retrieved evidence is combined with the specialized language model.
9. The system generates a grounded response with citations.
10. The response is evaluated and returned through the REST API to the existing ONSSA platform.

This workflow reflects the complete lifecycle of regulatory knowledge, from raw document acquisition to AI-assisted question answering, while preserving traceability, modularity, and institutional control.

Chapter 10 — Knowledge Ingestion Pipeline

10. Knowledge Ingestion Pipeline

10.1 Purpose

The Knowledge Ingestion Pipeline is the foundational component of the proposed architecture and constitutes the single entry point through which regulatory information enters the Artificial Intelligence system.

Its primary responsibility is to transform heterogeneous ONSSA regulatory documents into structured, validated, and reusable knowledge assets suitable for downstream Artificial Intelligence applications.

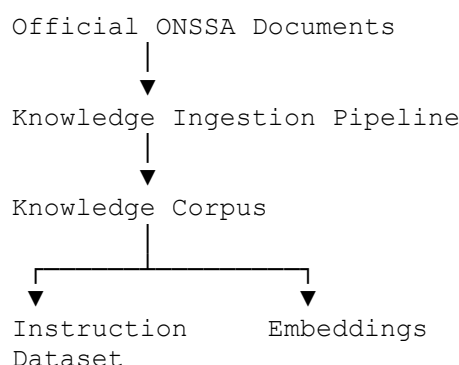
Unlike traditional document indexing systems, the pipeline does not simply extract text. It progressively enriches raw documents by preserving legal structure, contextual relationships, metadata, and traceability, ultimately producing a canonical Knowledge Corpus that serves as the single source of truth for the entire platform.

Every downstream component—including the Instruction Dataset, semantic embeddings, model specialization, Retrieval-Augmented Generation (RAG), and response generation—depends exclusively on the outputs produced by this pipeline.

Consequently, the quality, consistency, and reliability of the complete AI service are directly determined by the quality of the Knowledge Ingestion Pipeline.

10.2 Architectural Position

Within the overall system architecture, the Knowledge Ingestion Pipeline occupies the central position between raw documentary sources and Artificial Intelligence components.



The pipeline represents the only component authorized to transform raw regulatory documents into machine-readable knowledge.

No downstream component accesses original documents directly.

This architectural constraint guarantees consistency, reproducibility, and traceability across the complete system.

10.3 Design Principles

The Knowledge Ingestion Pipeline has been designed according to the following engineering principles.

Canonical Processing

Every document, regardless of its original format, follows the same logical processing workflow.

HTML pages, PDF files, Word documents, scanned documents, and official bulletins ultimately converge toward a common structured representation.

Single Processing Path

Each document is processed exactly once through the complete pipeline.

Once incorporated into the Knowledge Corpus, downstream components consume the structured representation rather than reprocessing the original document.

This approach eliminates duplicated processing logic and simplifies maintenance.

Incremental Enrichment

Information is progressively enriched throughout the pipeline.

Each processing stage adds new information without discarding validated outputs produced by previous stages.

For example:

- parsing extracts textual content;
 - metadata extraction enriches document descriptors;
 - legal structuring identifies regulatory hierarchy;
 - intelligent chunking creates semantically coherent retrieval units.
-

Traceability Preservation

Every generated artifact maintains a direct relationship with its originating document.

Traceability information is preserved throughout the complete processing workflow, allowing every response generated by the AI system to be linked back to its supporting official sources.

Idempotent Processing

Executing the pipeline multiple times on an unchanged document collection shall produce identical Knowledge Corpus artifacts.

This guarantees reproducibility, facilitates debugging, and simplifies future corpus regeneration.

10.4 Pipeline Inputs

The pipeline accepts heterogeneous documentary resources originating from official ONSSA repositories.

Supported input types include:

- HTML pages;
- PDF documents;
- Microsoft Word documents;

- Official Bulletins;
- Laws;
- Implementing Decrees;
- Ministerial Orders;
- Good Practice Guides;
- Administrative Forms;
- Images;
- Scanned Documents.

Each document enters the pipeline together with any available contextual information, including its source location, publication details, language, and document category.

10.5 Pipeline Outputs

The primary output of the pipeline is a structured Knowledge Corpus stored in JSON format.

This corpus contains machine-readable regulatory knowledge enriched with metadata and legal structure.

The Knowledge Corpus subsequently serves as the source for two additional knowledge assets:

- the Instruction Dataset used for QLoRA Fine-Tuning;
- the semantic Embedding Dataset used to populate the Vector Database.

No other component generates these assets independently.

10.6 Processing Stages

The pipeline consists of ten sequential processing stages.

Each stage has a clearly defined responsibility and produces validated outputs consumed by the subsequent stage.

Stage	Primary Responsibility	Output
Discovery	Identify eligible ONSSA resources	Document inventory
Download	Retrieve official documents	Local document repository
Parsing	Extract structured textual content	Normalized document
OCR	Recover text from scanned content	Searchable text

Stage	Primary Responsibility	Output
Cleaning	Remove noise and normalize content	Clean document
Deduplication	Detect identical or overlapping documents	Unique document set
Metadata Extraction	Enrich documents with descriptive attributes	Metadata records
Document Classification	Categorize regulatory documents	Classified corpus
Legal Structuring	Preserve legal hierarchy and document organization	Structured legal representation
Intelligent Chunking	Produce semantically coherent retrieval units	Knowledge Corpus

Each stage is described in detail in the following sections.

10.7 Stage Dependencies

The pipeline follows a strictly sequential dependency model.

A stage consumes only validated outputs generated by the immediately preceding stage.

For example:

```
Discovery
  ↓
Download
  ↓
Parsing
  ↓
OCR
  ↓
Cleaning
  ↓
Deduplication
  ↓
Metadata Extraction
  ↓
Document Classification
  ↓
Legal Structuring
  ↓
Intelligent Chunking
```

This deterministic execution model simplifies validation, debugging, testing, and maintenance.

10.8 Validation Gates

Before information progresses to the next processing stage, validation criteria shall be applied.

Typical validation activities include:

- document accessibility verification;
- parsing completeness;
- OCR confidence assessment;
- metadata consistency checks;
- duplicate detection validation;
- legal structure verification;
- chunk quality assessment.

Documents failing validation remain traceable and may be reprocessed after corrective actions.

This prevents low-quality information from propagating through the remainder of the AI pipeline.

10.9 Error Handling Strategy

The ingestion process is designed to be fault tolerant.

Failures affecting individual documents shall not interrupt processing of unrelated resources.

Typical recoverable errors include:

- inaccessible documents;
- corrupted PDF files;
- OCR failures;
- unsupported encodings;
- incomplete metadata.

Such failures shall be recorded, classified, and reported while allowing the remainder of the pipeline to continue processing successfully.

10.10 Pipeline Deliverables

Upon successful execution, the Knowledge Ingestion Pipeline produces the following deliverables:

- a validated Knowledge Corpus;
- complete document metadata;
- preserved legal hierarchy;
- semantic document chunks;
- full traceability to official ONSSA sources.

These deliverables form the foundation upon which all subsequent AI components are built.

No downstream processing—including Fine-Tuning, embedding generation, semantic retrieval, or response generation—may bypass or replace the outputs of the Knowledge Ingestion Pipeline.

Chapter 11 — Vertical Slice #1

11. Vertical Slice #1

11.1 Purpose

The implementation of a production-grade Artificial Intelligence system involves multiple interdependent components, including document acquisition, knowledge engineering, model specialization, semantic retrieval, inference, evaluation, deployment, and system integration.

Attempting to process the complete ONSSA documentary repository during the initial development phase would significantly increase technical complexity, prolong validation cycles, and make it difficult to isolate architectural issues.

To reduce implementation risk while validating the proposed architecture under realistic operating conditions, the project adopts a **Vertical Slice** development strategy.

Rather than maximizing document coverage, the first implementation aims to validate the complete end-to-end architecture on a representative subset of ONSSA's regulatory documentation.

The objective is to demonstrate that every architectural component functions correctly together before expanding the system to additional regulatory domains.

11.2 Vertical Slice Scope

The first Vertical Slice is intentionally restricted to the following regulatory hierarchy:

```

Regulation
├── Transversal Regulation
│   └── Food Safety

```

This subset contains representative examples of the document types and regulatory structures found throughout the ONSSA repository, including:

- Laws;
- Implementing Decrees;
- Ministerial Orders;
- Good Practice Guides;
- Official regulatory publications.

The selected scope is sufficiently representative to validate the complete AI workflow while remaining manageable during the initial implementation.

11.3 Why This Domain

The **Food Safety** regulatory domain was selected because it satisfies several critical engineering and business criteria.

First, it contains heterogeneous document formats representative of the broader ONSSA documentary repository, including structured web pages, PDF documents, legal texts, and technical guidance.

Second, the domain exhibits a well-defined legal hierarchy, allowing validation of legal structuring, metadata extraction, and citation mechanisms.

Third, it provides realistic regulatory questions that require combining information from multiple official sources, making it suitable for evaluating semantic retrieval and grounded response generation.

Finally, this domain represents a strategic regulatory area for ONSSA, making it an appropriate candidate for validating the practical value of the proposed solution.

11.4 Validation Objectives

The Vertical Slice is not intended to evaluate only the language model.

Instead, it validates the complete technical architecture.

Successful completion of the Vertical Slice demonstrates that the following capabilities operate correctly as an integrated system:

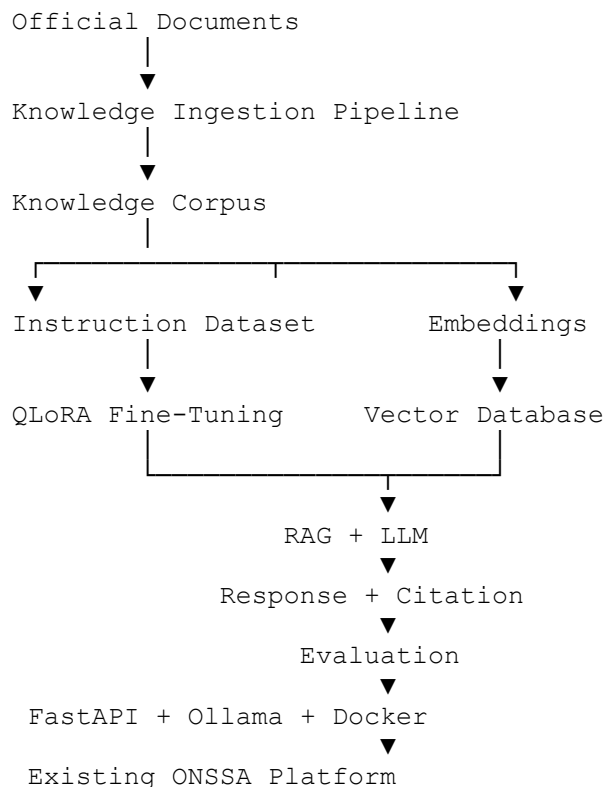
- document discovery;
- document acquisition;
- heterogeneous document processing;
- Knowledge Corpus generation;
- Instruction Dataset generation;

- embedding generation;
- model specialization;
- semantic retrieval;
- grounded response generation;
- citation generation;
- evaluation workflow;
- local deployment;
- integration with the existing ONSSA platform.

The Vertical Slice is therefore an architectural validation milestone rather than a functional prototype.

11.5 End-to-End Workflow Validation

The Vertical Slice validates the complete lifecycle of regulatory knowledge.



Each component must operate successfully before the architecture is considered validated.

11.6 Acceptance Criteria

The Vertical Slice shall be considered successful when all of the following conditions are satisfied:

Knowledge Engineering

- Official documents are successfully discovered and acquired.
- Supported document formats are processed correctly.
- OCR is applied where necessary.
- Metadata is extracted consistently.
- Legal structure is preserved.
- The Knowledge Corpus is successfully generated.

Model Preparation

- The Instruction Dataset is generated from the Knowledge Corpus.
- Semantic embeddings are generated successfully.
- The Vector Database is populated correctly.
- The language model is successfully specialized using QLoRA.

Inference

- User queries retrieve relevant regulatory passages.
- Retrieved evidence is incorporated into the model context.
- Responses are generated exclusively from retrieved documentation.
- Citations correctly reference supporting documents.

Deployment

- The complete AI service executes locally.
- REST API endpoints are operational.
- The service integrates successfully with the existing ONSSA platform.

11.7 Success Metrics

The objective of the Vertical Slice is architectural validation rather than maximum document coverage.

Success is measured according to the following qualitative outcomes:

- the complete ingestion pipeline executes successfully;
- the Knowledge Corpus accurately represents the processed documents;
- downstream AI assets are generated without manual intervention;
- retrieval consistently identifies relevant regulatory content;
- generated responses remain grounded in official documentation;
- citations provide sufficient traceability for verification;
- the AI service operates entirely within ONSSA's local infrastructure;
- the architecture supports future corpus expansion without redesign.

These outcomes demonstrate that the proposed architecture is technically sound and ready to be extended to additional regulatory domains.

11.8 Expansion Strategy

Once the Vertical Slice has been validated, expanding the system does not require architectural modifications.

Additional regulatory domains are incorporated by repeating the same ingestion workflow on new document collections.

The existing architecture—including the Knowledge Ingestion Pipeline, Knowledge Corpus, Instruction Dataset generation, embedding generation, model specialization, Retrieval-Augmented Generation workflow, evaluation framework, and deployment infrastructure—remains unchanged.

This incremental approach minimizes implementation risk, preserves architectural consistency, and enables the progressive transformation of the complete ONSSA documentary repository into a unified institutional knowledge system.

Chapter 12 — Knowledge Corpus Specification

12. Knowledge Corpus Specification

12.1 Purpose

The **Knowledge Corpus** is the canonical machine-readable representation of ONSSA's regulatory knowledge and constitutes the single source of truth for every Artificial Intelligence component within the proposed architecture.

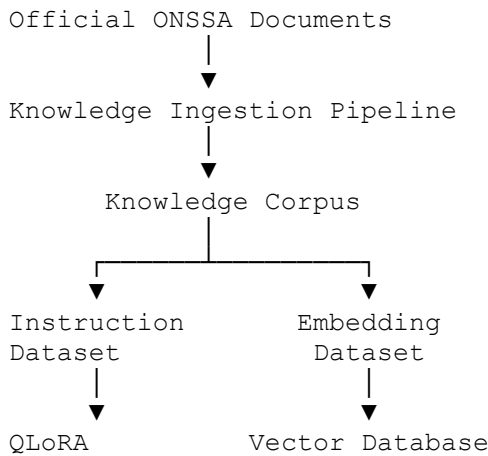
It is the primary deliverable produced by the Knowledge Ingestion Pipeline and represents the transition point between document engineering and AI-driven knowledge processing.

Unlike the original ONSSA documents, which exist in heterogeneous formats and structures, the Knowledge Corpus provides a unified, normalized, and structured representation of regulatory information while preserving legal semantics, document hierarchy, metadata, and traceability.

No downstream component—including Fine-Tuning, embedding generation, Retrieval-Augmented Generation (RAG), or response generation—consumes raw documents directly. All AI components operate exclusively on the Knowledge Corpus.

12.2 Architectural Role

Within the overall system architecture, the Knowledge Corpus occupies a central position.



The Knowledge Corpus decouples document processing from Artificial Intelligence processing.

This separation ensures that document engineering is performed once, while multiple downstream AI workflows reuse the same validated knowledge base.

12.3 Design Objectives

The Knowledge Corpus has been designed to satisfy the following objectives.

Canonical Representation

All ONSSA regulatory documents shall be transformed into a common representation regardless of their original format.

HTML pages, PDF documents, Word files, scanned documents, and Official Bulletins shall all produce structurally consistent corpus entries.

Knowledge Reusability

The same corpus shall simultaneously support:

- model specialization;
- semantic retrieval;
- future AI services;
- semantic search;
- knowledge management;

- corpus analytics.

Knowledge extraction is therefore performed once and reused across multiple downstream applications.

Traceability

Every knowledge element contained within the corpus shall remain traceable to its originating official document.

Traceability shall never be lost throughout the processing pipeline.

Each chunk, section, and metadata record must preserve a reference to its original source.

Legal Integrity

The corpus shall preserve the legal organization of regulatory documents.

The ingestion process shall not flatten or destroy the legal hierarchy established by the original publications.

Articles, chapters, sections, titles, annexes, and document relationships shall remain identifiable after processing.

Version Independence

The corpus shall remain independent from any specific language model, embedding model, or vector database implementation.

Future replacement of downstream AI technologies shall not require regeneration of the Knowledge Corpus unless the source documents themselves change.

12.4 Logical Structure

The Knowledge Corpus is organized as a collection of structured regulatory documents.

Each document consists of three complementary information layers.

Document Layer

Describes the regulatory document itself.

Typical information includes:

- unique document identifier;
 - document title;
 - document type;
 - publication date;
 - language;
 - source URL;
 - official reference;
 - document version;
 - processing status.
-

Structural Layer

Represents the internal legal organization of the document.

Examples include:

- chapters;
- sections;
- articles;
- paragraphs;
- annexes;
- tables;
- legal references.

This layer preserves the original legal hierarchy required for accurate citation and contextual retrieval.

Semantic Layer

Represents the smallest retrievable knowledge units generated during intelligent chunking.

Each semantic unit contains:

- normalized textual content;
- associated metadata;
- legal position within the document;
- traceability information;
- chunk identifier.

These semantic units constitute the primary retrieval objects used by the RAG system.

12.5 Corpus Metadata

Each document stored in the Knowledge Corpus shall include standardized metadata.

Mandatory metadata categories include:

Identification Metadata

- Document Identifier
 - Original File Identifier
 - Source URL
 - File Type
 - Processing Timestamp
-

Regulatory Metadata

- Document Type
 - Regulatory Domain
 - Regulatory Subdomain
 - Legal Authority
 - Publication Reference
 - Official Bulletin Reference
-

Linguistic Metadata

- Language
 - Character Encoding
 - OCR Status
-

Processing Metadata

- Parser Used
- Processing Version
- Chunking Version
- Validation Status

Standardized metadata enables filtering, retrieval, auditing, regeneration, and citation generation.

12.6 Corpus Granularity

The corpus preserves knowledge at multiple levels of granularity.

These include:

- document level;
- chapter level;
- section level;
- article level;
- paragraph level;
- semantic chunk level.

Maintaining multiple granularities improves retrieval precision while preserving regulatory context.

12.7 Corpus Quality Requirements

Before inclusion in the Knowledge Corpus, processed documents shall satisfy defined quality criteria.

Examples include:

- successful parsing;
- readable textual content;
- valid metadata;
- preserved legal hierarchy;
- successful duplicate detection;
- acceptable OCR quality;
- successful semantic segmentation.

Documents failing validation shall be excluded from downstream AI processing until corrected.

12.8 Downstream Consumers

The Knowledge Corpus serves as the exclusive input for multiple architectural components.

These include:

[Instruction Dataset Generator](#)

Transforms corpus knowledge into supervised instruction-response examples suitable for QLoRA Fine-Tuning.

Embedding Generator

Converts semantic chunks into dense vector representations stored within the Vector Database.

Evaluation Framework

Uses the corpus as the reference dataset for validating retrieval quality, citation correctness, and grounding accuracy.

Future AI Services

The architecture intentionally positions the Knowledge Corpus as a reusable institutional asset capable of supporting additional AI applications beyond regulatory question answering.

12.9 Versioning Strategy

The Knowledge Corpus shall support explicit versioning.

Each corpus generation shall be uniquely identifiable and reproducible.

Versioning enables:

- corpus regeneration;
- rollback;
- auditability;
- comparison between corpus generations;
- reproducible model training;
- reproducible embedding generation.

Every downstream artifact shall reference the exact corpus version from which it was generated.

12.10 Summary

The Knowledge Corpus represents the central knowledge asset of the proposed architecture.

It establishes a clear separation between document engineering and Artificial Intelligence processing while preserving legal integrity, traceability, metadata consistency, and long-term maintainability.

By serving as the single authoritative knowledge source for Fine-Tuning, semantic retrieval, evaluation, and future AI capabilities, the Knowledge Corpus provides the stable foundation upon which the entire ONSSA regulatory intelligence service is built.

Chapter 13 — Instruction Dataset Specification

13. Instruction Dataset Specification

13.1 Purpose

The **Instruction Dataset** is a supervised training dataset derived exclusively from the Knowledge Corpus and dedicated to the specialization of the Large Language Model through parameter-efficient fine-tuning.

Its objective is **not** to transfer ONSSA's regulatory knowledge into the model parameters.

Instead, the dataset teaches the model **how to behave** when interacting with ONSSA regulatory information.

This includes:

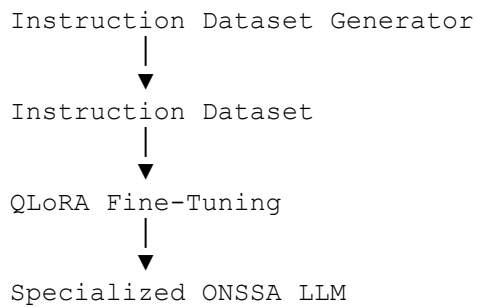
- understanding regulatory questions;
- adopting ONSSA terminology;
- following institutional communication standards;
- structuring responses consistently;
- citing official sources correctly;
- refusing unsupported requests;
- using retrieved evidence appropriately.

The Instruction Dataset therefore specializes the model's behavior while leaving factual regulatory knowledge external to the model.

13.2 Architectural Position

Within the proposed architecture, the Instruction Dataset is generated after the Knowledge Corpus has been validated.

Knowledge Corpus
↓
▼



The dataset is never created directly from raw documents.

All training examples originate from the structured Knowledge Corpus, ensuring consistency, traceability, and reproducibility.

13.3 Design Objectives

The Instruction Dataset has been designed to achieve the following objectives.

Behavioral Specialization

The dataset teaches the model the expected interaction patterns rather than the complete regulatory corpus.

The model learns:

- response organization;
- professional writing style;
- ONSSA-specific vocabulary;
- citation conventions;
- instruction compliance;
- response boundaries.

Separation Between Knowledge and Behavior

A fundamental architectural principle of this project is the separation between **knowledge** and **behavior**.

Regulatory knowledge remains external to the model and is retrieved dynamically through the RAG pipeline.

Behavioral knowledge is learned during Fine-Tuning.

This separation offers several advantages:

- easier regulatory updates;

- smaller Fine-Tuning datasets;
- reduced risk of outdated knowledge;
- lower retraining costs;
- improved maintainability.

Reproducibility

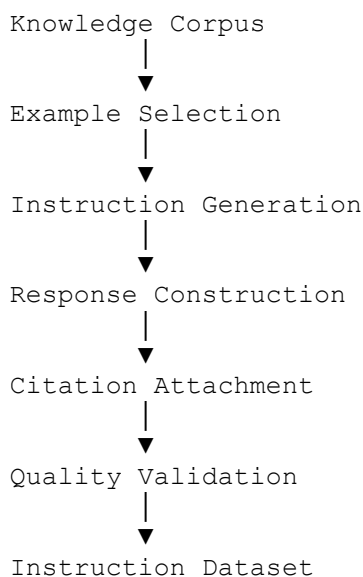
The dataset shall be generated deterministically from a specific version of the Knowledge Corpus.

Given the same corpus version and generation configuration, the resulting Instruction Dataset shall be reproducible.

13.4 Dataset Generation Workflow

The Instruction Dataset is automatically derived from the Knowledge Corpus through a dedicated generation process.

The workflow follows these stages:



The generation process is fully automated to facilitate future corpus regeneration.

13.5 Dataset Content

Each training example represents a complete interaction between a user and the AI assistant.

A training example contains four logical components.

User Instruction

A natural-language regulatory question representing the type of request expected from ONSSA users.

Examples include:

- requests for regulatory clarification;
 - document identification;
 - procedural questions;
 - legal requirement inquiries;
 - compliance-related questions.
-

Context

The supporting regulatory information extracted from the Knowledge Corpus.

The context represents the evidence available to the model during supervised learning.

Expected Response

The expected response demonstrates:

- factual grounding;
 - professional language;
 - coherent organization;
 - regulatory terminology;
 - concise explanation;
 - instruction compliance.
-

Citation Information

Training examples explicitly demonstrate how official regulatory references shall appear in generated responses.

The objective is to teach citation behavior rather than memorizing specific legal texts.

13.6 Supported Instruction Categories

The dataset shall include a diverse range of interaction patterns representative of future production usage.

Examples include:

Information Retrieval

Questions requesting specific regulatory information.

Regulatory Explanation

Questions asking for clarification of regulatory concepts.

Procedure Guidance

Questions concerning administrative or regulatory procedures described within official documents.

Document Navigation

Questions asking where particular information can be found within ONSSA documentation.

Comparative Questions

Questions requiring synthesis of multiple retrieved regulatory sources.

Insufficient Evidence

Examples demonstrating the correct behavior when retrieved evidence is insufficient to answer reliably.

These examples are essential for reducing hallucinations during production inference.

13.7 Quality Requirements

Every training example shall satisfy defined quality criteria before inclusion in the dataset.

These include:

Grounding

Responses must be supported by information contained within the Knowledge Corpus.

Consistency

Identical regulatory situations shall produce consistent response structures.

Citation Correctness

Citation formatting shall follow institutional conventions.

Language Quality

Responses shall be grammatically correct, professionally written, and aligned with ONSSA terminology.

Instruction Compliance

Examples shall demonstrate the expected behavior for following user instructions while respecting system limitations.

13.8 Excluded Content

The Instruction Dataset intentionally excludes several categories of information.

These include:

- unsupported regulatory claims;
- speculative interpretations;
- invented citations;
- unofficial regulatory sources;
- contradictory examples;
- duplicated interactions.

The dataset shall also avoid encouraging the model to answer questions outside the available documentary evidence.

13.9 Relationship with the Knowledge Corpus

The Instruction Dataset is not an independent knowledge source.

Instead, it represents a behavioral projection of the Knowledge Corpus.

This relationship establishes several important constraints:

- the corpus always precedes the dataset;
- dataset regeneration follows corpus updates;
- corpus version changes trigger dataset version updates;
- every training example remains traceable to the originating corpus version.

This dependency guarantees consistency between the regulatory knowledge base and the specialized language model.

13.10 Deliverables

Successful execution of the Instruction Dataset generation process produces:

- a validated supervised training dataset;
- standardized instruction-response examples;
- consistent citation demonstrations;
- reproducible dataset versions;
- metadata linking every example to the corresponding Knowledge Corpus version.

These deliverables constitute the direct input for the QLoRA Fine-Tuning process described in the next chapter.

13.11 Summary

The Instruction Dataset enables behavioral specialization of the language model without embedding the complete ONSSA regulatory corpus into the model parameters.

By separating institutional behavior from regulatory knowledge, the proposed architecture maintains a clear distinction between what the model **knows how to do** and what it **knows**.

This design supports scalable knowledge updates, reduces retraining requirements, and ensures that authoritative regulatory information continues to originate exclusively from the Knowledge Corpus during Retrieval-Augmented Generation.

Chapter 14 — Fine-Tuning Strategy

14. Fine-Tuning Strategy

14.1 Purpose

The objective of the Fine-Tuning process is to adapt a locally deployed open-source Large Language Model to the communication standards, terminology, and interaction patterns required for ONSSA's regulatory environment.

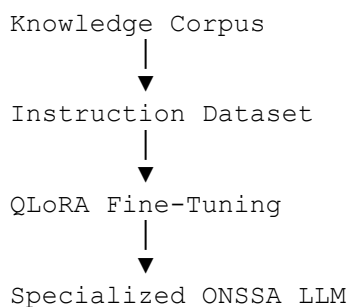
Fine-Tuning is **not** intended to encode the complete regulatory corpus into the model parameters.

Instead, it specializes the model's behavior so that it can effectively interpret retrieved regulatory evidence, communicate using institutional language, follow predefined response conventions, and consistently generate evidence-based answers.

This architectural decision preserves a clear separation between **behavioral adaptation** and **knowledge retrieval**, improving maintainability, scalability, and long-term sustainability.

14.2 Architectural Position

Within the proposed architecture, Fine-Tuning consumes only the Instruction Dataset generated from the Knowledge Corpus.



The model never accesses raw ONSSA documents during the Fine-Tuning process.

Regulatory documents are transformed into behavioral examples before training begins.

14.3 Design Philosophy

The Fine-Tuning strategy follows a fundamental design principle:

Teach the model how to respond, not what the regulations are.

The distinction is essential.

The model is expected to learn:

- institutional terminology;
- response organization;
- regulatory writing style;
- citation conventions;
- interaction boundaries;
- instruction-following behavior.

Conversely, the model is **not** expected to memorize:

- legal articles;
- regulatory texts;
- decrees;
- laws;
- official procedures;
- numerical regulatory values.

These remain external to the model and are retrieved dynamically through the RAG pipeline.

14.4 Why Behavioral Fine-Tuning Instead of Knowledge Fine-Tuning

Encoding the entire ONSSA regulatory repository into model parameters would introduce several technical and operational drawbacks.

Regulatory Evolution

ONSSA regulations evolve over time through new laws, amendments, implementing decrees, and updated technical guidance.

If regulatory knowledge were embedded directly into model parameters, each significant regulatory update would require retraining the model.

By externalizing knowledge into the Knowledge Corpus, updates primarily require rerunning the Knowledge Ingestion Pipeline and regenerating the vector index, while the Fine-Tuned model remains reusable.

Explainability

Knowledge retrieved from the Knowledge Corpus remains directly traceable to official documents.

If knowledge were stored only in model parameters, it would become difficult to justify generated answers or identify the supporting legal references.

The proposed architecture preserves complete traceability from generated responses back to the originating regulatory documents.

Maintainability

Behavioral Fine-Tuning results in smaller, more stable training datasets.

The model behavior evolves slowly compared with regulatory content, reducing retraining frequency and simplifying long-term maintenance.

Reduced Hallucination Risk

Because factual information is retrieved during inference rather than memorized during training, the model is less likely to produce outdated or unsupported regulatory statements.

Grounded generation therefore improves overall system reliability.

14.5 QLoRA as the Fine-Tuning Method

The proposed architecture adopts **Quantized Low-Rank Adaptation (QLoRA)** as the Fine-Tuning strategy.

QLoRA offers several advantages for institutional deployments.

Parameter Efficiency

Only a small subset of trainable parameters is updated.

This significantly reduces computational requirements while preserving model performance.

Reduced Hardware Requirements

QLoRA enables specialization of modern Large Language Models using substantially less GPU memory than full Fine-Tuning.

This makes the approach suitable for deployment within ONSSA's local infrastructure.

Preservation of Base Model Knowledge

The underlying general-purpose reasoning capabilities of the base model remain largely intact.

The Fine-Tuning process introduces ONSSA-specific behavior without replacing the model's general language understanding.

Reproducibility

QLoRA adapters can be versioned independently from the base model.

This simplifies experimentation, rollback, deployment, and long-term maintenance.

14.6 Fine-Tuning Inputs

The Fine-Tuning process consumes the following inputs:

- validated Instruction Dataset;
- selected base Large Language Model;
- Fine-Tuning configuration;
- training hyperparameters;
- evaluation configuration.

No raw regulatory documents participate directly in model training.

14.7 Fine-Tuning Outputs

Successful execution of the Fine-Tuning process produces:

- specialized ONSSA adapter weights;
- training metrics;
- validation metrics;

- model version metadata;
- reproducible training configuration.

The specialized model remains compatible with the Retrieval-Augmented Generation architecture.

14.8 Training Scope

The Fine-Tuning process teaches the model to:

- understand ONSSA terminology;
- structure regulatory responses;
- produce professional institutional language;
- integrate retrieved evidence coherently;
- generate citations consistently;
- refuse unsupported requests;
- communicate uncertainty appropriately.

The process intentionally excludes memorization of regulatory facts.

14.9 Evaluation During Fine-Tuning

Model specialization shall be evaluated continuously throughout training.

Evaluation focuses primarily on behavioral quality rather than factual memorization.

Representative evaluation dimensions include:

- instruction-following capability;
- response consistency;
- citation formatting;
- language quality;
- response organization;
- robustness against unsupported requests.

The resulting model is accepted only if it satisfies predefined behavioral quality criteria.

14.10 Relationship with Retrieval-Augmented Generation

The Fine-Tuned model is not intended to operate independently.

Instead, it forms one component of the Retrieval-Augmented Generation architecture.

During production inference:

1. the retrieval system identifies relevant regulatory evidence;
2. retrieved passages are supplied to the model;
3. the Fine-Tuned model interprets this evidence;
4. a grounded response is generated;
5. supporting citations are returned.

This interaction clearly separates knowledge retrieval from language generation.

14.11 Versioning

Each Fine-Tuned model shall be uniquely associated with:

- a base model version;
- a Knowledge Corpus version;
- an Instruction Dataset version;
- a training configuration version;
- a QLoRA adapter version.

This versioning strategy ensures full reproducibility and simplifies future maintenance.

14.12 Summary

The proposed Fine-Tuning strategy specializes the behavior of a Large Language Model rather than embedding ONSSA's regulatory knowledge within its parameters.

By combining behavioral specialization through QLoRA with knowledge retrieval through the Knowledge Corpus and Vector Database, the architecture achieves a clear separation of concerns.

This approach reduces retraining costs, simplifies regulatory updates, preserves explainability, minimizes hallucinations, and establishes a sustainable foundation for long-term operation within ONSSA's local infrastructure.

The Fine-Tuned model therefore acts as an intelligent regulatory communicator, while authoritative knowledge continues to originate exclusively from the Knowledge Corpus through the Retrieval-Augmented Generation workflow.

Chapter 15 — Retrieval-Augmented Generation (RAG) Architecture

15. Retrieval-Augmented Generation (RAG) Architecture

15.1 Purpose

The Retrieval-Augmented Generation (RAG) architecture constitutes the inference engine of the proposed ONSSA Regulatory Intelligence Service.

Its responsibility is to transform a user's regulatory question into a grounded, explainable, and evidence-based response by combining semantic retrieval with a specialized Large Language Model.

Unlike conventional conversational AI systems that rely primarily on knowledge stored within model parameters, the proposed architecture separates **knowledge retrieval** from **language generation**.

This separation ensures that regulatory knowledge remains external, verifiable, continuously maintainable, and directly traceable to official ONSSA publications.

During inference, the Large Language Model does not generate responses from memory alone. Instead, it interprets documentary evidence retrieved from the Knowledge Corpus and synthesizes this evidence into a coherent response that follows ONSSA's communication standards.

The RAG architecture therefore represents the operational bridge between institutional knowledge and natural language interaction.

15.2 Architectural Objectives

The Retrieval-Augmented Generation architecture has been designed to satisfy several strategic objectives.

Evidence-Based Generation

Every response shall originate from documentary evidence retrieved from the Knowledge Corpus.

The language model shall never act as the primary source of regulatory information.

Instead, it functions as an interpreter of retrieved evidence.

Explainability

Every generated statement should remain traceable to one or more official regulatory documents.

This traceability enables users to verify regulatory information independently.

Explainability is therefore an inherent property of the architecture rather than an optional feature.

Regulatory Freshness

Because regulatory information is retrieved dynamically during inference, updates to the regulatory repository become available without retraining the language model.

Updating the Knowledge Corpus and rebuilding the Vector Database are sufficient to expose new regulations to the assistant.

Reduced Hallucination Risk

Grounding the language model with retrieved documentary evidence significantly reduces unsupported generation.

When relevant evidence cannot be retrieved, the assistant shall explicitly communicate that the available regulatory information is insufficient rather than speculate.

Separation of Responsibilities

The architecture deliberately assigns distinct responsibilities to different components.

Component	Responsibility
Knowledge Corpus	Stores validated regulatory knowledge
Embedding Dataset	Represents semantic meaning
Vector Database	Retrieves relevant information
Specialized LLM	Interprets retrieved evidence

Component	Responsibility
Citation Generator	Produces documentary references

This separation improves maintainability and facilitates independent evolution of each component.

15.3 Why Retrieval-Augmented Generation

Several architectural alternatives were evaluated before selecting the proposed Retrieval-Augmented Generation approach.

Alternative 1 — Full Knowledge Fine-Tuning

One possible strategy consists of fine-tuning the language model using the complete regulatory corpus so that the model memorizes ONSSA regulations.

Although conceptually simple, this approach presents several limitations.

Regulatory updates require model retraining.

Explainability is significantly reduced because factual knowledge resides within model parameters.

Verifying generated responses becomes more difficult.

Model size and training costs increase substantially.

Consequently, this approach was rejected.

Alternative 2 — Prompt-Only Retrieval

Another possibility is to rely exclusively on retrieved documents without model specialization.

Although this simplifies training, the resulting system often exhibits inconsistent terminology, variable response structures, inconsistent citation formatting, and weaker instruction-following behavior.

The resulting user experience does not satisfy institutional communication requirements.

Selected Approach

The proposed solution combines both strategies.

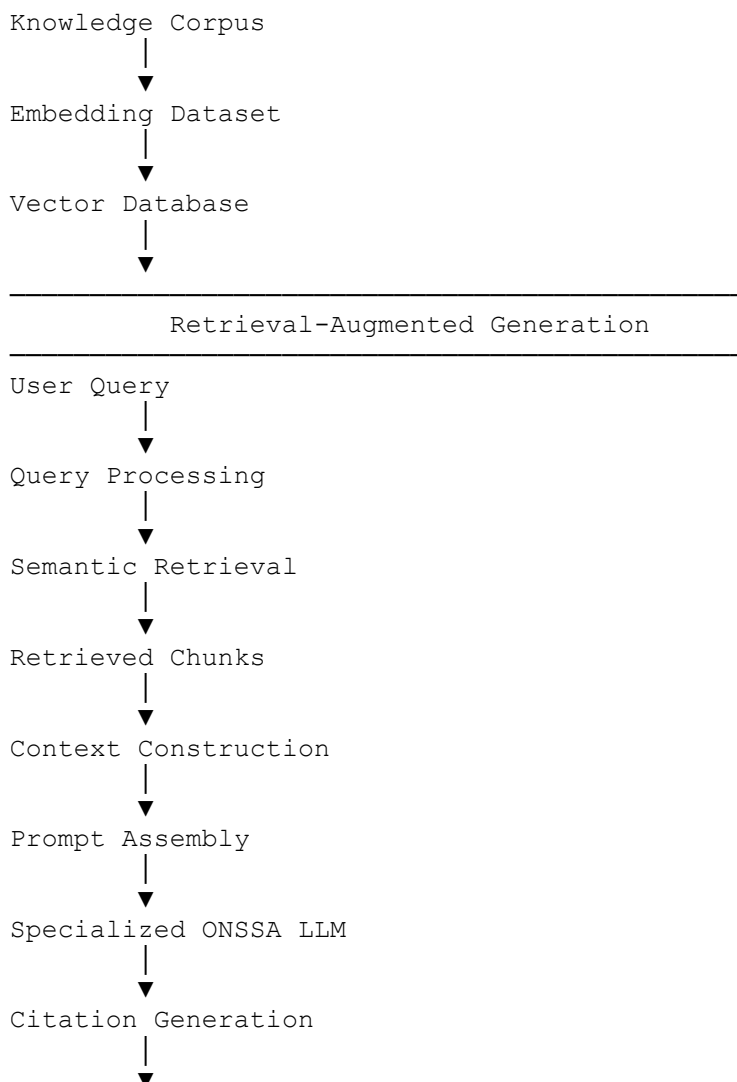
The Knowledge Corpus supplies factual knowledge through semantic retrieval.

The specialized language model contributes reasoning, language understanding, institutional terminology, and response organization.

This hybrid architecture combines the strengths of Retrieval-Augmented Generation and parameter-efficient Fine-Tuning while minimizing their respective limitations.

15.4 Architectural Position

Within the global architecture, the Retrieval-Augmented Generation layer connects semantic retrieval with natural language generation.





This logical decomposition illustrates the internal workflow of the RAG component while remaining fully consistent with the high-level architecture defined in Chapter 9.

15.5 End-to-End Inference Workflow

The inference workflow follows a deterministic sequence of operations.

1. A user submits a regulatory question through the existing ONSSA platform.
2. The REST API receives the request and forwards it to the Retrieval-Augmented Generation engine.
3. The user query is analyzed and prepared for semantic retrieval.
4. A vector representation of the query is generated using the same embedding model employed during corpus indexing.
5. The Vector Database performs semantic similarity search over the indexed Knowledge Corpus.
6. The most relevant knowledge chunks are returned together with their associated metadata.
7. Retrieved evidence is validated and organized.
8. A structured context is assembled.
9. The specialized language model receives:
 - the user question;
 - the retrieved evidence;
 - system instructions;
 - citation metadata.
10. The language model synthesizes a grounded response.
11. Documentary citations are attached.
12. The response is returned to the existing ONSSA platform through the REST API.

Every response generated by the system follows this same execution path.

15.6 Internal Components

The RAG architecture is composed of several cooperating components.

Although these components execute as part of a unified inference workflow, each has a clearly defined responsibility.

Query Processing

Responsible for preparing incoming user requests before semantic retrieval.

Typical responsibilities include:

- language normalization;
- whitespace normalization;
- encoding validation;
- query preprocessing;
- request validation.

The objective is to provide a clean and consistent representation of the user's request to downstream retrieval components.

Query Embedding

The processed query is converted into a dense semantic vector using the embedding model associated with the current Knowledge Corpus.

This representation captures semantic similarity rather than keyword overlap.

Using the same embedding model for both indexing and querying ensures consistency between stored knowledge and user requests.

Semantic Retrieval

The semantic retrieval component searches the Vector Database to identify document chunks that are most relevant to the user's question.

Unlike traditional keyword search, semantic retrieval retrieves information based on conceptual similarity.

The retrieved chunks constitute the factual evidence used during response generation.

Context Builder

Retrieved chunks are not forwarded directly to the language model.

Instead, they are organized into a structured inference context.

The Context Builder is responsible for:

- ordering retrieved evidence;
- removing redundant information;

- preserving document relationships;
- maintaining legal hierarchy;
- associating metadata with each retrieved chunk.

The resulting context maximizes relevance while minimizing unnecessary prompt length.

Prompt Assembly

The Prompt Assembly component combines multiple information sources into a single inference prompt.

The assembled prompt includes:

- system instructions;
- user question;
- retrieved regulatory evidence;
- citation metadata;
- behavioral guidance inherited from the Fine-Tuned model.

This structured prompt forms the complete input supplied to the specialized language model.

Specialized Language Model

The Fine-Tuned language model interprets the assembled context and produces a coherent regulatory response.

The model is responsible for:

- synthesizing retrieved evidence;
- maintaining professional language;
- applying ONSSA terminology;
- following institutional formatting conventions;
- expressing uncertainty when evidence is insufficient.

The model is not responsible for retrieving regulatory information.

Knowledge retrieval has already been completed by previous components.

Citation Generator

After response generation, citation information associated with retrieved chunks is incorporated into the final answer.

The Citation Generator preserves traceability between generated statements and official regulatory sources.

Whenever possible, citations reference:

- document title;
- legal reference;
- article number;
- section identifier;
- originating publication.

This mechanism enables users to verify every response using official documentation.

15.7 Context Management

Purpose

The quality of a Retrieval-Augmented Generation system depends not only on retrieving relevant information but also on presenting that information to the language model in a coherent and structured manner.

The Context Management component is responsible for transforming the retrieved document chunks into a context that maximizes factual accuracy while remaining within the language model's context window.

Rather than forwarding retrieved chunks exactly as they are stored in the Vector Database, the system constructs a structured context optimized for inference.

Context Construction Principles

The context construction process follows several guiding principles.

Relevance First

Only document fragments directly related to the user's question shall be included.

Irrelevant passages increase prompt size, consume context window capacity, and may reduce response quality.

Preservation of Legal Context

Individual chunks shall not lose their legal meaning.

Whenever necessary, surrounding contextual information shall accompany the retrieved passage to preserve the interpretation of legal provisions.

For example, an article referring to conditions defined in a previous section should be presented together with the necessary contextual elements.

Hierarchical Ordering

Retrieved evidence shall be organized according to the legal organization of the originating documents rather than the retrieval order alone.

When multiple documents contribute to the response, the context shall preserve:

- document hierarchy;
- chapter hierarchy;
- section hierarchy;
- article sequence.

Maintaining this organization improves the model's ability to synthesize regulatory information consistently.

Redundancy Elimination

Semantic retrieval may return overlapping chunks.

The Context Builder shall identify redundant information and avoid inserting duplicate evidence into the prompt.

This optimization reduces unnecessary token consumption while preserving informational completeness.

Metadata Preservation

Every retrieved chunk shall retain its associated metadata throughout context construction.

Metadata required during inference includes:

- document identifier;
- document title;
- document type;
- legal reference;
- publication reference;
- language;
- chunk identifier.

This metadata is subsequently consumed by the Citation Generator.

15.8 Prompt Engineering Strategy

Purpose

Prompt construction defines how retrieved evidence is presented to the specialized language model.

The objective is to provide the model with sufficient information to generate an accurate, well-structured, and evidence-based response without encouraging unsupported inference.

Prompt Components

Each inference prompt consists of four logical sections.

System Instructions

The system instructions define the operational behavior of the assistant.

Typical responsibilities include:

- defining the assistant's role;
- enforcing institutional communication standards;
- prohibiting unsupported claims;
- requiring evidence-based responses;
- specifying citation requirements.

These instructions remain stable across user requests.

User Query

The original user question is inserted without altering its semantic intent.

This preserves the user's information need while allowing the model to interpret the request in conjunction with the retrieved regulatory evidence.

Retrieved Evidence

Retrieved document passages constitute the factual foundation of the response.

Evidence is inserted in a structured format that preserves:

- source document;
- legal hierarchy;
- chunk boundaries;

- associated metadata.

The language model is expected to prioritize this evidence when generating its response.

Response Instructions

Additional guidance specifies the expected response format.

Typical instructions include:

- answer only using retrieved evidence;
 - clearly state when information is insufficient;
 - avoid speculation;
 - maintain professional language;
 - include documentary citations.
-

15.9 Citation Strategy

Purpose

Citation generation is a fundamental architectural requirement rather than a presentation feature.

Every generated response shall allow users to verify the underlying regulatory sources independently.

The Citation Strategy therefore guarantees transparency, explainability, and institutional trust.

Citation Sources

Citations originate exclusively from metadata preserved throughout the Knowledge Ingestion Pipeline.

The language model does not invent citation information.

Instead, citation metadata accompanies retrieved chunks throughout the complete inference workflow.

Citation Elements

Whenever available, citations should include:

- document title;
- regulatory document type;
- article number;
- section identifier;
- publication reference;
- Official Bulletin reference.

The exact presentation format may evolve without modifying the underlying architecture.

Citation Consistency

Responses synthesizing information from multiple regulatory documents shall preserve the relationship between statements and their supporting sources.

Citation consistency is particularly important when multiple legal texts jointly answer a user's question.

15.10 Failure Handling

Purpose

Production systems must anticipate situations in which retrieval or generation cannot proceed as expected.

The Retrieval-Augmented Generation architecture therefore incorporates explicit failure-handling mechanisms.

Failures shall be handled predictably, transparently, and without compromising regulatory reliability.

No Relevant Evidence Retrieved

If semantic retrieval fails to identify sufficiently relevant regulatory information, the assistant shall not generate speculative answers.

Instead, it shall inform the user that no reliable information could be identified within the available ONSSA documentation.

Conflicting Evidence

In cases where multiple retrieved documents appear to provide inconsistent information, the assistant shall avoid selecting one interpretation without justification.

Whenever possible, the response shall:

- identify the relevant documents;
 - describe the apparent inconsistency;
 - encourage consultation of the official regulatory sources.
-

Incomplete Context

Some regulatory provisions depend on external references or related legal texts.

If the retrieved context is insufficient to support a reliable conclusion, the assistant shall explicitly indicate this limitation.

Internal Processing Errors

Unexpected failures affecting retrieval, prompt construction, or inference shall be reported through standardized system logs.

The user-facing response shall remain informative while avoiding exposure of internal implementation details.

15.11 Hallucination Mitigation

Design Objective

Reducing hallucinations represents one of the primary objectives of the proposed architecture.

Rather than relying on a single mitigation technique, the system combines multiple complementary mechanisms throughout the inference pipeline.

Knowledge Externalization

Regulatory knowledge remains outside the language model.

This minimizes the risk of outdated or fabricated regulatory statements.

Evidence-Based Prompting

The language model receives validated documentary evidence before response generation.

This encourages grounded reasoning rather than unsupported completion.

Behavioral Fine-Tuning

The specialized model is explicitly trained to refuse unsupported requests and acknowledge uncertainty when evidence is unavailable.

Mandatory Citations

Every response includes documentary references.

The obligation to produce citations reinforces evidence-based generation and facilitates manual verification.

Continuous Evaluation

Hallucination rates shall be monitored through the Evaluation Framework described in Chapter 16.

Observed hallucinations shall guide future improvements to retrieval quality, corpus construction, and model specialization.

15.12 Retrieval Quality Optimization

Purpose

The effectiveness of the Retrieval-Augmented Generation architecture depends directly on the quality of the retrieved evidence.

Retrieval optimization therefore focuses on maximizing the relevance, completeness, and precision of the documentary context supplied to the language model.

Unlike traditional search engines that primarily optimize keyword matching, the proposed system optimizes semantic relevance while preserving regulatory context.

Retrieval Precision

The retrieval engine shall prioritize document chunks that are semantically aligned with the user's regulatory question.

High precision reduces irrelevant context, improves response quality, and minimizes unnecessary token consumption during inference.

Retrieval Recall

The retrieval process shall identify all regulatory passages necessary to answer the user's question.

When a regulatory concept spans multiple documents or legal provisions, retrieval shall return sufficient evidence to support a complete and accurate response.

Context Completeness

Retrieval quality is evaluated not only by the relevance of individual chunks but also by the completeness of the assembled context.

A complete context should provide the language model with all essential information required to formulate a grounded answer.

Metadata-Aware Retrieval

Document metadata preserved within the Knowledge Corpus shall remain available throughout the retrieval process.

Metadata enables future enhancements such as:

- document-type filtering;
- language-specific retrieval;
- regulatory-domain filtering;
- publication-date constraints;
- document version filtering.

Although the initial Vertical Slice focuses on semantic retrieval, the architecture is designed to support these additional retrieval dimensions without structural modifications.

15.13 Performance Considerations

Design Objectives

The RAG architecture shall provide interactive response times while maintaining high retrieval accuracy and response reliability.

Performance optimization shall never compromise regulatory correctness or traceability.

Efficient Vector Search

Semantic retrieval shall be optimized to minimize query latency even as the Knowledge Corpus expands to additional ONSSA regulatory domains.

The Vector Database shall support efficient similarity search over a growing collection of semantic document chunks.

Prompt Efficiency

The Context Builder shall optimize prompt size by including only evidence necessary for answering the user's request.

Reducing unnecessary context improves inference efficiency and allows more effective use of the language model's context window.

Modular Scalability

Each inference component shall scale independently.

Future optimization of one component, such as retrieval or prompt construction, shall not require redesign of unrelated architectural layers.

Local Execution

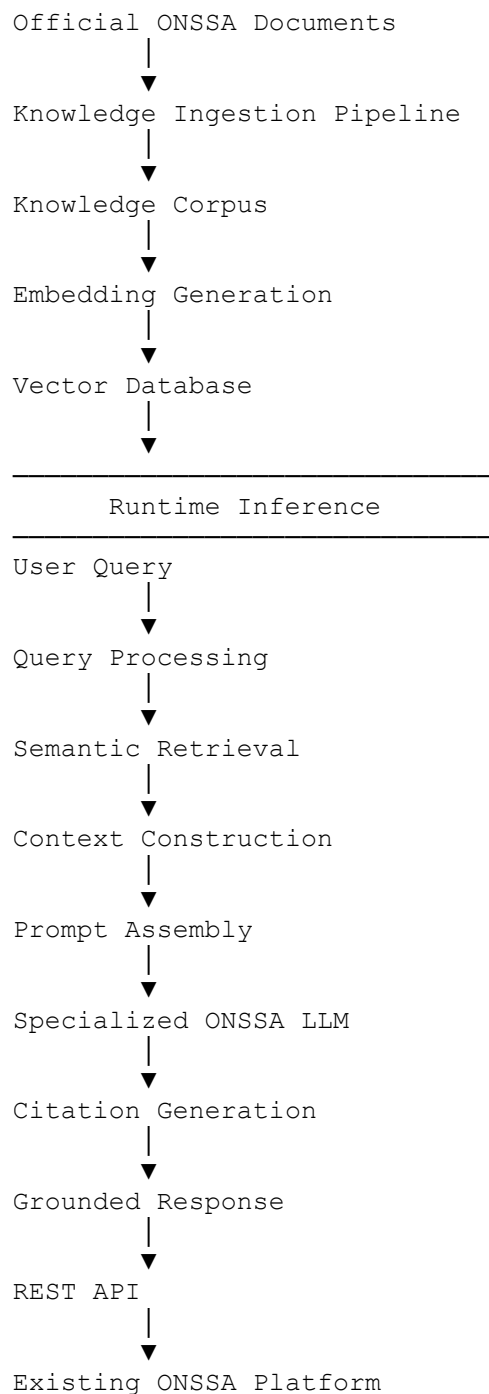
All inference operations—including embedding generation, semantic retrieval, language model inference, and citation generation—shall execute within ONSSA's local infrastructure.

This guarantees compliance with the project's confidentiality and data sovereignty requirements.

15.14 Operational RAG Lifecycle

The Retrieval-Augmented Generation architecture participates in a broader operational lifecycle extending beyond individual inference requests.

The lifecycle begins with regulatory document acquisition and concludes with the delivery of grounded responses to end users.

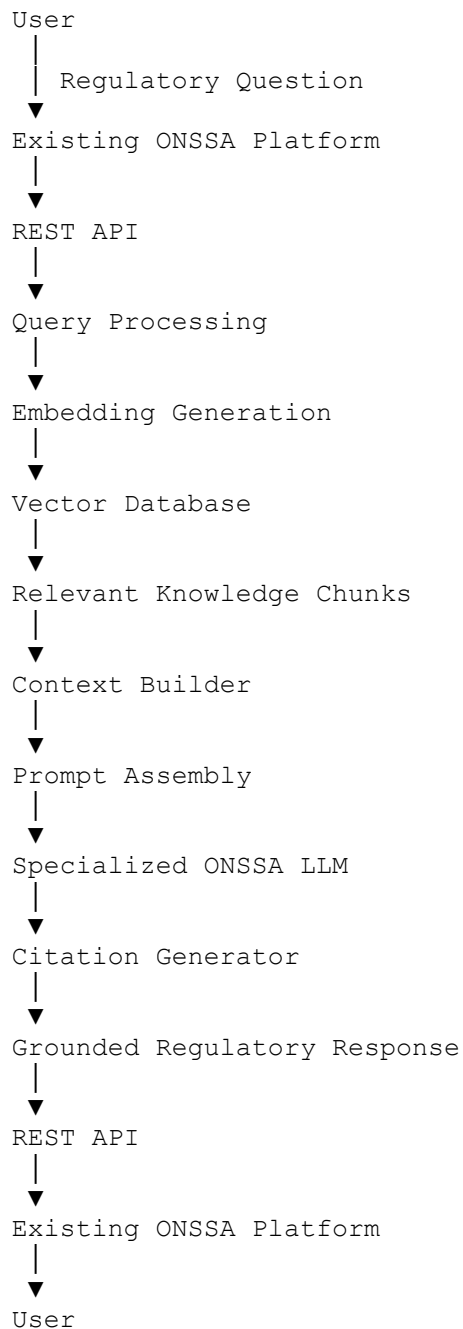


This lifecycle illustrates the interaction between offline knowledge engineering activities and online inference operations.

The separation of these phases enables efficient maintenance and supports future expansion of the regulatory repository.

15.15 End-to-End Sequence

The following sequence summarizes the interactions among the principal architectural components during a typical inference request.



This deterministic sequence guarantees that every response follows the same controlled inference workflow and preserves complete traceability from user query to supporting regulatory evidence.

15.16 Architectural Benefits

The proposed Retrieval-Augmented Generation architecture provides several long-term advantages for ONSSA.

Regulatory Maintainability

Updates to regulations primarily require regeneration of the Knowledge Corpus and Vector Database rather than retraining the language model.

This significantly reduces maintenance costs and accelerates regulatory updates.

Explainability

Every response remains linked to official ONSSA documentation through mandatory citation generation.

Users can independently verify the legal basis of each answer.

Scalability

The architecture supports gradual expansion from the initial Vertical Slice to the complete ONSSA documentary repository.

New regulatory domains are incorporated by extending the Knowledge Corpus without redesigning the inference architecture.

Component Independence

Knowledge engineering, retrieval, language generation, evaluation, and deployment remain logically independent.

This modularity simplifies future improvements and reduces coupling between architectural components.

Institutional Sustainability

The proposed architecture establishes a reusable institutional knowledge infrastructure capable of supporting future AI services beyond regulatory question answering.

The Knowledge Corpus becomes a long-term organizational asset rather than a project-specific artifact.

15.17 Summary

The Retrieval-Augmented Generation architecture constitutes the operational core of the proposed ONSSA Regulatory Intelligence Service.

By combining semantic retrieval with a behaviorally specialized Large Language Model, the system delivers grounded, explainable, and evidence-based regulatory responses while maintaining a strict separation between knowledge management and language generation.

This architecture ensures that regulatory knowledge remains centralized within the Knowledge Corpus, retrieval remains responsible for factual evidence, and the language model focuses on interpretation, communication, and response synthesis.

The resulting system achieves high maintainability, regulatory traceability, reduced hallucination risk, and long-term adaptability while operating entirely within ONSSA's local infrastructure.

The following chapter defines the Evaluation Framework used to measure the quality, reliability, and operational readiness of the complete AI service.

Chapter 16 — Evaluation Framework

16. Evaluation Framework

16.1 Purpose

The Evaluation Framework establishes the methodology used to measure, validate, and continuously improve the overall quality of the ONSSA Regulatory Intelligence Service.

Evaluation is not limited to the Large Language Model. Instead, it encompasses every major component of the architecture, including document processing, knowledge construction, semantic retrieval, language generation, citation generation, API behavior, and system integration.

The framework ensures that the AI service satisfies functional, technical, and operational requirements before deployment within ONSSA's production environment.

16.2 Evaluation Objectives

The Evaluation Framework has five primary objectives:

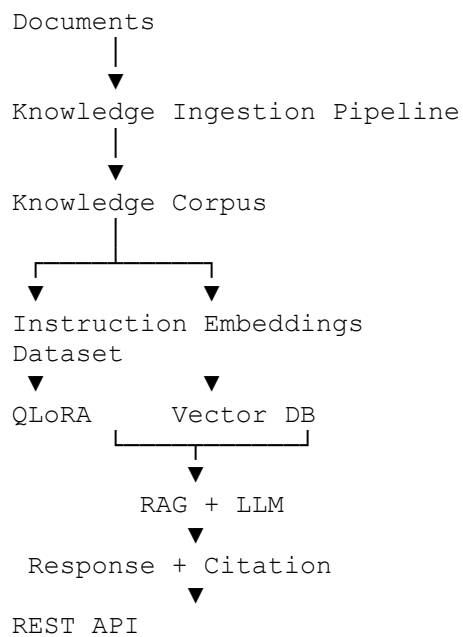
- verify the correctness of the Knowledge Corpus;

- assess the quality of semantic retrieval;
- measure the quality of generated responses;
- evaluate citation accuracy and traceability;
- validate production readiness.

Together, these objectives provide an end-to-end assessment of the complete AI workflow.

16.3 Evaluation Scope

Evaluation shall cover the entire architecture.



Each architectural layer shall be evaluated independently and as part of the complete end-to-end workflow.

16.4 Knowledge Ingestion Evaluation

Purpose

The first evaluation stage verifies that the Knowledge Ingestion Pipeline correctly transforms raw ONSSA documents into a structured Knowledge Corpus.

Evaluation focuses on the integrity and consistency of the generated knowledge assets.

Evaluation Criteria

The following aspects shall be assessed:

- successful document discovery;
- successful document acquisition;
- parsing completeness;
- OCR quality;
- metadata completeness;
- duplicate detection effectiveness;
- preservation of legal hierarchy;
- chunking consistency;
- corpus validation.

Any detected issues shall be corrected before downstream AI components are regenerated.

16.5 Knowledge Corpus Evaluation

The generated Knowledge Corpus shall be evaluated according to several quality dimensions.

Structural Integrity

Verification that document hierarchy has been preserved.

Metadata Completeness

Verification that mandatory metadata fields are present and consistent.

Traceability

Verification that every semantic chunk remains linked to its originating official document.

Consistency

Verification that corpus generation is deterministic and reproducible.

16.6 Retrieval Evaluation

Purpose

Retrieval quality directly determines the quality of generated responses.

The retrieval component shall therefore be evaluated independently from the language model.

Retrieval Accuracy

Evaluation shall determine whether relevant regulatory documents are successfully retrieved for representative user questions.

Retrieval Precision

Retrieved evidence should be highly relevant to the user's question.

Reducing irrelevant context improves response quality and inference efficiency.

Retrieval Recall

The retrieval process should identify all significant regulatory passages required to answer the user's request.

Context Completeness

The assembled inference context shall contain sufficient evidence for grounded response generation.

16.7 Language Model Evaluation

The specialized language model shall be evaluated primarily on behavioral quality rather than factual memorization.

Representative evaluation dimensions include:

- instruction following;
- institutional terminology;
- language quality;
- response organization;
- clarity;
- consistency;

- uncertainty communication.

Behavioral evaluation complements retrieval evaluation rather than replacing it.

16.8 Response Evaluation

Generated responses shall be evaluated using multiple qualitative criteria.

Grounding

Every factual statement should originate from retrieved regulatory evidence.

Correctness

Responses shall accurately represent the information contained within official ONSSA documentation.

Completeness

Responses should address all aspects of the user's request whenever sufficient evidence exists.

Clarity

Responses shall remain understandable, professionally written, and appropriately structured.

Consistency

Equivalent regulatory questions should produce consistent answers across repeated executions.

16.9 Citation Evaluation

Citation quality represents a critical success criterion.

Evaluation shall verify:

- citation presence;
- citation completeness;

- citation correctness;
- document traceability;
- consistency between generated statements and supporting references.

Responses lacking appropriate citations shall be considered incomplete.

16.10 Hallucination Evaluation

The architecture explicitly seeks to minimize unsupported generation.

Evaluation shall therefore identify responses containing:

- fabricated regulations;
- invented legal references;
- unsupported factual claims;
- speculative interpretations.

Observed hallucinations shall be analyzed to determine whether the underlying cause originates from:

- retrieval failure;
- corpus deficiencies;
- prompt construction;
- model behavior.

This analysis guides targeted improvements rather than indiscriminate retraining.

16.11 End-to-End Evaluation

Individual component validation is insufficient to guarantee production readiness.

The complete workflow shall therefore be evaluated from user query to final response.

Representative scenarios include:

- regulatory information requests;
- legal clarification;
- procedural guidance;
- document identification;
- cross-document reasoning;
- unsupported requests.

Successful completion of these scenarios demonstrates the operational readiness of the complete architecture.

16.12 Human Evaluation

Although automated evaluation provides objective measurements, institutional validation remains essential.

Subject matter experts shall review representative responses to assess:

- regulatory correctness;
- legal interpretation;
- terminology;
- citation relevance;
- practical usefulness.

Expert feedback shall be incorporated into future iterations of the Knowledge Corpus, Instruction Dataset, and retrieval configuration.

16.13 Continuous Improvement

Evaluation is not a one-time activity.

Results shall continuously inform improvements to:

- document processing;
- corpus construction;
- metadata extraction;
- chunking strategy;
- retrieval configuration;
- Fine-Tuning dataset;
- prompt engineering.

This continuous feedback loop ensures progressive enhancement of the AI service over time.

16.14 Acceptance Criteria

The AI service shall be considered ready for production when:

- the Knowledge Corpus satisfies all validation requirements;
- retrieval consistently identifies relevant evidence;
- responses remain grounded in official documentation;
- citations are complete and accurate;
- hallucinations remain within acceptable limits;
- REST API endpoints operate reliably;

- end-to-end workflows satisfy institutional requirements.

Production deployment shall occur only after these acceptance criteria have been validated.

16.15 Summary

The Evaluation Framework provides a comprehensive methodology for assessing every architectural layer of the ONSSA Regulatory Intelligence Service.

By evaluating document engineering, semantic retrieval, language generation, citation quality, and operational behavior as an integrated system, the framework ensures that the deployed AI service remains reliable, explainable, maintainable, and aligned with ONSSA's institutional requirements.

The following chapter describes the deployment architecture that operationalizes the validated AI service within ONSSA's local infrastructure.

Chapter 17 — Deployment Architecture

17. Deployment Architecture

17.1 Purpose

The Deployment Architecture defines how the ONSSA Regulatory Intelligence Service is packaged, deployed, operated, and maintained within ONSSA's local infrastructure.

Its objective is to transform the validated Artificial Intelligence architecture into a production-ready service while preserving security, maintainability, reproducibility, and institutional data sovereignty.

The deployment architecture is designed to operate entirely on-premises without relying on external Artificial Intelligence providers or cloud-based inference services.

17.2 Deployment Principles

The deployment strategy follows several guiding principles.

Local Execution

All system components shall execute within ONSSA's infrastructure.

This includes:

- document processing;
- Knowledge Corpus generation;
- embedding generation;
- vector indexing;
- language model inference;
- REST API services.

No production data shall leave ONSSA's infrastructure.

Modular Services

Each major architectural component shall be deployed as an independent service whenever practical.

This modular organization simplifies maintenance, testing, upgrades, and future scalability.

Containerization

Production services shall be packaged as Docker containers.

Containerization provides:

- reproducible deployments;
- environment consistency;
- simplified dependency management;
- portability across compatible infrastructures;
- streamlined updates and rollback procedures.

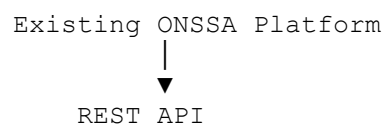
Infrastructure Independence

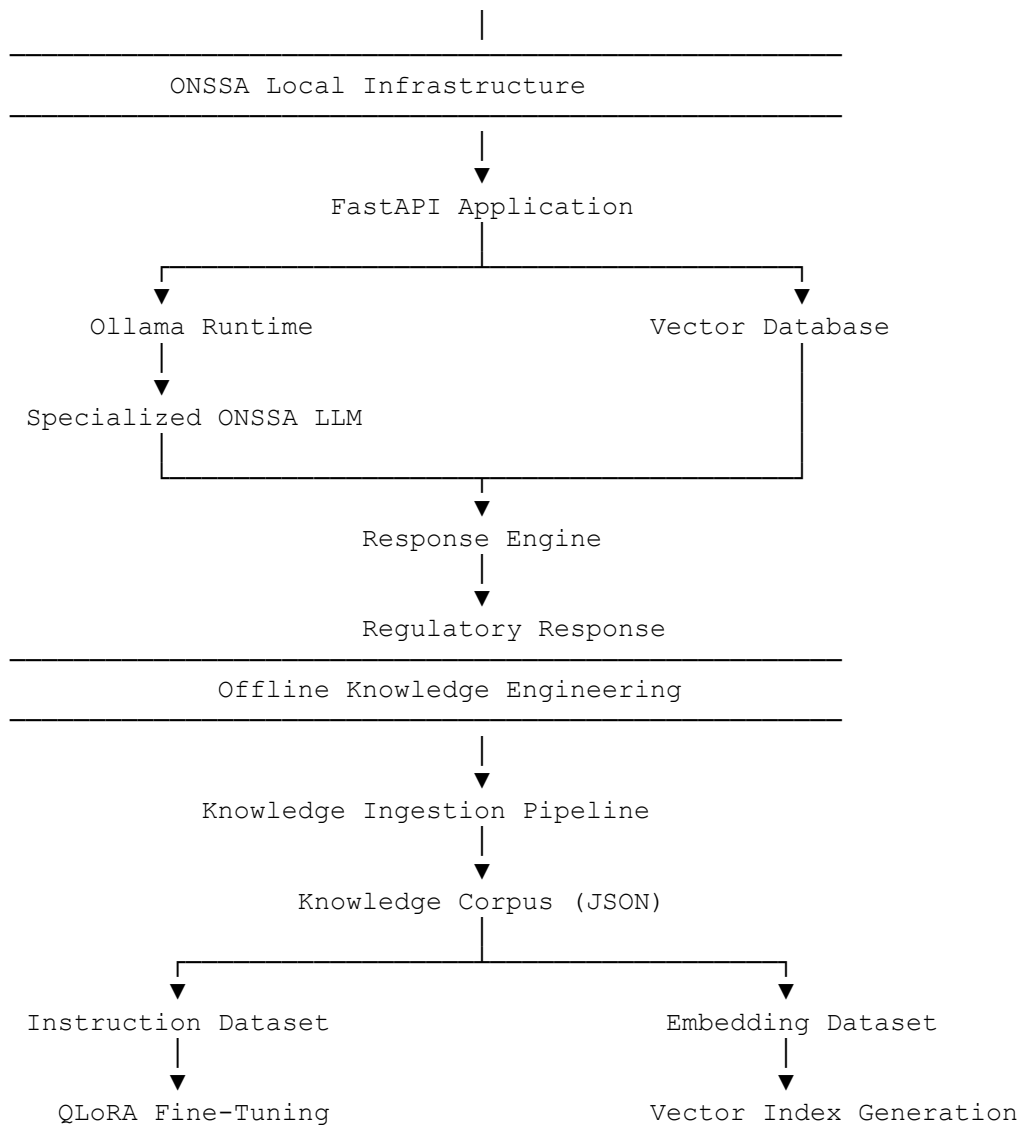
The AI service shall remain independent from the ONSSA presentation layer.

Communication between the existing ONSSA platform and the AI service shall occur exclusively through standardized REST interfaces.

17.3 High-Level Deployment Architecture

The deployment architecture is illustrated below.





The architecture clearly separates offline knowledge engineering from online inference services.

17.4 Deployment Layers

Layer 1 — Knowledge Engineering Environment

This environment executes the offline data engineering workflow.

Its responsibilities include:

- document discovery;
- document acquisition;
- parsing;
- OCR;
- metadata extraction;

- legal structuring;
- chunk generation;
- Knowledge Corpus construction.

This layer is executed only when the regulatory corpus requires updates.

Layer 2 — Model Preparation Environment

The model preparation environment generates AI assets.

Responsibilities include:

- Instruction Dataset generation;
- embedding generation;
- QLoRA Fine-Tuning;
- vector index construction.

These activities occur offline and do not directly impact production inference.

Layer 3 — Production Inference Environment

The production environment hosts the AI service consumed by ONSSA users.

Core responsibilities include:

- REST API services;
- semantic retrieval;
- prompt construction;
- language model inference;
- citation generation;
- response delivery.

This layer handles all real-time user interactions.

17.5 FastAPI Service

FastAPI serves as the primary application layer of the AI service.

Its responsibilities include:

- receiving user requests;
- validating incoming payloads;
- orchestrating retrieval and inference;
- formatting responses;

- exposing REST endpoints;
- returning citations alongside generated responses.

FastAPI does not perform language model inference directly.

Instead, it coordinates the interaction between the retrieval system and the inference runtime.

17.6 Ollama Runtime

Ollama provides the local inference runtime for the specialized language model.

Its responsibilities include:

- loading the specialized model;
- managing inference execution;
- processing prompts;
- generating responses.

The runtime remains entirely local and requires no external network connectivity during production inference.

17.7 Vector Database

The Vector Database stores semantic representations generated from the Knowledge Corpus.

Its responsibilities include:

- indexing semantic chunks;
- performing similarity search;
- returning relevant document fragments;
- supporting retrieval operations.

The Vector Database is updated only after regeneration of the Knowledge Corpus and Embedding Dataset.

17.8 Deployment Workflow

Deployment follows a structured sequence.

Phase 1 — Knowledge Engineering

The Knowledge Ingestion Pipeline processes official ONSSA documents and generates the Knowledge Corpus.

Phase 2 — AI Asset Generation

The Knowledge Corpus is transformed into:

- Instruction Dataset;
- Embedding Dataset.

The Instruction Dataset is used for QLoRA Fine-Tuning, while the Embedding Dataset populates the Vector Database.

Phase 3 — Production Deployment

The specialized language model, Vector Database, and FastAPI service are deployed within Docker containers.

After deployment, the AI service becomes available through the REST API.

17.9 Update Workflow

The deployment architecture supports incremental regulatory updates.

When ONSSA publishes new regulatory documents, the update workflow consists of:

1. rerunning the Knowledge Ingestion Pipeline;
2. regenerating the Knowledge Corpus;
3. regenerating semantic embeddings;
4. rebuilding the Vector Database.

Fine-Tuning is repeated only when changes affect the model's expected behavior, communication style, or instruction-following requirements.

Routine regulatory updates do not require retraining the model.

17.10 Security Considerations

The deployment architecture has been designed to satisfy institutional security requirements.

Key principles include:

- local document storage;
- local model execution;
- local vector indexing;
- no external AI APIs;
- secure REST communication;
- controlled access to AI services.

These principles preserve confidentiality and support ONSSA's data governance policies.

17.11 Operational Monitoring

The production environment shall support operational monitoring of key services.

Representative monitoring indicators include:

- service availability;
- API latency;
- inference latency;
- retrieval latency;
- CPU utilization;
- memory utilization;
- storage utilization;
- error rates.

Monitoring data shall support preventive maintenance and rapid incident response.

17.12 Backup and Recovery

The deployment architecture shall support backup and recovery of critical assets.

Recoverable components include:

- Knowledge Corpus;
- Instruction Dataset;
- Embedding Dataset;
- Vector Database;
- QLoRA adapter weights;
- deployment configuration;
- application configuration.

Recovery procedures shall minimize service interruption while preserving corpus integrity and model reproducibility.

17.13 Scalability

The deployment architecture supports progressive expansion without redesign.

Future scalability may include:

- larger regulatory corpora;
- additional language models;
- increased user concurrency;
- additional AI services;
- expanded regulatory domains.

The modular deployment model enables independent scaling of individual components according to operational requirements.

17.14 Summary

The Deployment Architecture operationalizes the complete ONSSA Regulatory Intelligence Service within ONSSA's local infrastructure.

By separating offline knowledge engineering from online inference, adopting containerized deployment, and exposing standardized REST interfaces through FastAPI, the architecture delivers a secure, maintainable, and scalable AI service fully aligned with the project's data sovereignty and institutional integration requirements.

The next chapter describes how this AI service integrates with the existing ONSSA platform while preserving the current application architecture and user experience.

Chapter 18 — Integration with ONSSA Platform

18. Integration with ONSSA Platform

18.1 Purpose

The objective of this chapter is to define how the ONSSA Regulatory Intelligence Service integrates with the existing ONSSA information system.

The project does **not** include the development of a new user platform or document management system.

Instead, it delivers an independent Artificial Intelligence service that exposes standardized REST interfaces and is consumed by the existing ONSSA platform.

This integration approach minimizes disruption to the current application ecosystem while enabling progressive adoption of AI-assisted regulatory search and question answering.

18.2 Integration Principles

The integration strategy follows five fundamental principles.

Service-Oriented Integration

The Artificial Intelligence solution is deployed as an independent service.

The existing ONSSA platform remains responsible for:

- user authentication;
- user interface;
- navigation;
- session management;
- business workflows.

The AI service is responsible exclusively for regulatory question answering.

Loose Coupling

The AI service shall remain loosely coupled to the existing platform.

Communication occurs only through documented REST APIs.

Neither system shall require knowledge of the other's internal implementation.

This separation simplifies maintenance and future evolution.

Backward Compatibility

The integration shall preserve existing ONSSA functionalities.

The introduction of the AI service shall not require redesign of the current application architecture.

Users who do not use AI-assisted search shall continue to use the platform without modification.

Incremental Adoption

The architecture supports gradual deployment.

The first production deployment targets the validated Vertical Slice covering:

Regulation → Transversal Regulation → Food Safety

After successful validation, additional regulatory domains may be integrated without architectural changes.

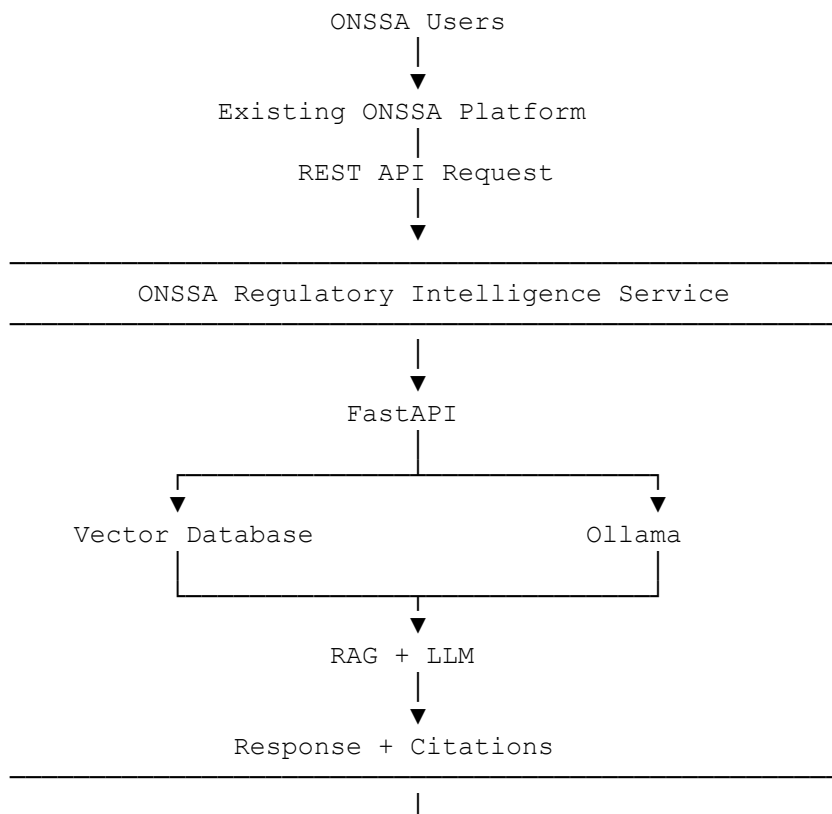
Local Communication

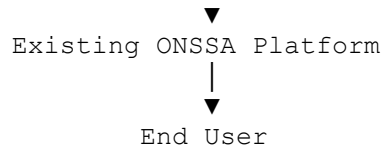
All communication between the ONSSA platform and the AI service shall occur within ONSSA's internal infrastructure.

No production requests are transmitted to external Artificial Intelligence services.

18.3 Integration Architecture

The interaction between the existing ONSSA platform and the AI service is illustrated below.





This architecture preserves the existing application while introducing AI capabilities as an independent backend service.

18.4 Integration Workflow

A typical user interaction follows the sequence below.

Step 1 — User Request

A user submits a regulatory question using the existing ONSSA interface.

The user experience remains unchanged except for the availability of the AI assistant.

Step 2 — API Invocation

The existing platform sends the request to the AI service through the REST API.

The transmitted payload contains the user's question and any contextual information required for processing.

Step 3 — AI Processing

The AI service executes the complete inference workflow:

1. query preprocessing;
 2. semantic retrieval;
 3. context construction;
 4. prompt assembly;
 5. language model inference;
 6. citation generation.
-

Step 4 — Response Delivery

The generated response and supporting citations are returned to the existing ONSSA platform.

The existing platform renders the response using its own presentation layer.

The AI service remains independent of any user interface implementation.

18.5 API Contract

The Artificial Intelligence service exposes REST endpoints designed for integration with the existing platform.

The API follows the following principles:

- stateless communication;
- JSON request and response payloads;
- standardized HTTP status codes;
- explicit error reporting;
- versioned endpoints;
- consistent response structures.

The detailed API specification is provided in the appendices.

18.6 Request Processing

For each incoming request, the AI service performs the following operations:

1. validate the request;
2. verify mandatory fields;
3. preprocess the query;
4. execute semantic retrieval;
5. invoke the specialized language model;
6. generate citations;
7. construct the response payload;
8. return the result to the calling application.

This deterministic workflow ensures predictable system behavior.

18.7 Response Structure

Every successful response shall contain:

- generated regulatory answer;

- supporting citations;
- referenced documents;
- processing metadata;
- response status.

When no reliable regulatory evidence is available, the response shall explicitly indicate this condition instead of generating speculative information.

18.8 Error Handling

The integration layer shall provide standardized error handling.

Representative error categories include:

Invalid Request

The request payload does not satisfy API requirements.

Processing Error

An internal processing failure prevents successful completion of the inference workflow.

Knowledge Not Found

The retrieval process cannot identify sufficiently relevant regulatory evidence.

Service Unavailable

The AI service is temporarily unavailable.

Error responses shall remain machine-readable and suitable for integration with the existing ONSSA application.

18.9 Security

Integration shall comply with ONSSA's institutional security requirements.

The AI service shall support:

- authenticated access;
- encrypted communication;
- request validation;
- controlled network exposure;
- audit logging.

Authentication remains under the responsibility of the existing ONSSA platform unless future architectural requirements specify otherwise.

18.10 Logging and Auditability

The AI service shall generate operational logs for system monitoring and troubleshooting.

Representative logged information includes:

- request identifier;
- processing timestamps;
- inference duration;
- retrieval duration;
- response status;
- error category.

Sensitive user information shall not be recorded unnecessarily.

Logging shall comply with institutional governance and data protection policies.

18.11 Future Integration Opportunities

Although the initial deployment focuses on regulatory question answering, the service-oriented architecture enables future integration with additional ONSSA applications.

Potential extensions include:

- intelligent document search;
- regulatory recommendation services;
- compliance assistance;
- semantic document navigation;
- internal knowledge support tools;
- decision support services.

These future capabilities can reuse the same Knowledge Corpus and Retrieval-Augmented Generation infrastructure.

18.12 Benefits of the Integration Strategy

The selected integration approach provides several advantages.

Minimal Impact

Existing applications remain largely unchanged.

Maintainability

The AI service evolves independently from the existing platform.

Scalability

Additional AI capabilities may be introduced without redesigning the existing information system.

Security

All processing remains within ONSSA's infrastructure.

Reusability

The same AI service may support multiple institutional applications.

18.13 Summary

The proposed integration architecture introduces the ONSSA Regulatory Intelligence Service as an independent Artificial Intelligence backend seamlessly connected to the existing ONSSA platform through standardized REST APIs.

This service-oriented approach preserves the current application ecosystem while enabling the progressive adoption of AI-powered regulatory assistance.

By maintaining clear separation between presentation, business logic, and Artificial Intelligence, the architecture supports long-term maintainability, institutional scalability, and secure on-premises operation without disrupting existing ONSSA workflows.

Chapter 19 — Development Roadmap

19.1 Purpose

The Development Roadmap defines the implementation strategy for the ONSSA Regulatory Intelligence Service. Rather than attempting to deliver the complete solution in a single iteration, the project follows an incremental development approach centered on a functional Vertical Slice. This methodology reduces technical risk, enables early validation, and facilitates continuous integration of additional regulatory domains.

19.2 Development Strategy

The project follows a phased implementation approach.

Phase 1 — Vertical Slice

Implement the complete end-to-end workflow for the initial regulatory scope:

- Regulation
- Transversal Regulation
- Food Safety

Deliverables:

- Knowledge Ingestion Pipeline
- Knowledge Corpus
- Embedding generation
- Vector Database
- Initial RAG pipeline
- Functional REST API
- Integration prototype

This phase validates the overall architecture before expanding the corpus.

Phase 2 — Knowledge Engineering

Enhance the ingestion pipeline to improve:

- parsing robustness;
- OCR accuracy;
- metadata extraction;
- legal structuring;
- intelligent chunking;
- corpus validation.

Deliverable:

A production-ready Knowledge Corpus generation pipeline.

Phase 3 — Model Specialization

Generate the Instruction Dataset and execute QLoRA Fine-Tuning.

Deliverables:

- Instruction Dataset
 - Specialized ONSSA model
 - Adapter weights
 - Model evaluation report
-

Phase 4 — Retrieval Optimization

Improve retrieval quality through:

- embedding optimization;
- retrieval tuning;
- prompt engineering;
- citation validation.

Deliverable:

Optimized Retrieval-Augmented Generation pipeline.

Phase 5 — System Validation

Perform comprehensive evaluation of:

- ingestion quality;
- retrieval performance;
- response quality;
- citation correctness;
- end-to-end workflows.

Deliverable:

System validation report.

Phase 6 — Production Deployment

Deploy the AI service within ONSSA's infrastructure using Docker, FastAPI, and Ollama.

Deliverables:

- production deployment;
 - operational documentation;
 - integration with the existing ONSSA platform.
-

19.3 Milestones

Milestone	Expected Outcome
M1	Vertical Slice operational
M2	Knowledge Corpus validated
M3	Specialized LLM available
M4	RAG pipeline optimized
M5	Evaluation completed
M6	Production deployment

19.4 Summary

The phased roadmap minimizes implementation risk while ensuring that each architectural component is validated before extending the system to additional regulatory domains.

Chapter 20 — Risks & Mitigation

20.1 Purpose

This chapter identifies the principal technical and operational risks associated with the project and defines mitigation strategies to reduce their impact.

20.2 Risk Assessment

Risk	Impact	Mitigation
Poor document quality	Incorrect corpus generation	OCR validation, manual verification, preprocessing
Scanned or low-quality PDFs	Information loss	OCR pipeline with quality control
Duplicate documents	Retrieval inconsistencies	Automatic deduplication during ingestion
Regulatory updates	Outdated responses	Periodic corpus regeneration and vector index rebuild
Hallucinations	Incorrect regulatory guidance	Evidence-based RAG, mandatory citations, behavioral fine-tuning
Retrieval failures	Incomplete answers	Retrieval optimization and evaluation
Infrastructure limitations	Increased latency	QLoRA, quantized models, efficient vector indexing
Integration issues	Deployment delays	REST API versioning and integration testing
Security vulnerabilities	Data exposure	On-premises deployment, authentication, encrypted communication
User adoption	Low utilization	Progressive deployment and expert feedback

20.3 Risk Management Strategy

Risk management is based on four principles:

- early detection through continuous evaluation;
 - modular architecture to isolate failures;
 - incremental deployment using the Vertical Slice;
 - continuous monitoring after production deployment.
-

20.4 Summary

The proposed architecture minimizes project risks through modular design, local deployment, evidence-based response generation, and continuous validation.

Chapter 21 — Future Work

21.1 Purpose

The proposed architecture has been intentionally designed to support future evolution without requiring major architectural redesign.

21.2 Planned Extensions

Future developments may include:

- extension to the complete ONSSA regulatory repository;
 - support for additional document categories;
 - multilingual retrieval optimization;
 - hybrid search combining semantic and lexical retrieval;
 - document re-ranking;
 - automated corpus synchronization;
 - user feedback integration;
 - administrative dashboard;
 - analytics and monitoring tools;
 - support for additional institutional AI services.
-

21.3 Long-Term Vision

The long-term objective is to transform the AI service into a reusable institutional knowledge platform capable of supporting multiple ONSSA applications beyond regulatory question answering.

The Knowledge Corpus becomes a strategic organizational asset reusable across future Artificial Intelligence initiatives.

21.4 Summary

The proposed architecture provides a scalable foundation for future AI capabilities while preserving compatibility with the existing ONSSA information system.

Chapter 22 — Appendices

Appendix A — Glossary

Term	Definition
Knowledge Corpus	Canonical structured representation of ONSSA regulatory knowledge
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
QLoRA	Quantized Low-Rank Adaptation
Embeddings	Dense semantic vector representations
Vector Database	Database storing semantic embeddings for retrieval
Chunk	Smallest retrievable semantic unit
OCR	Optical Character Recognition
REST API	Interface exposing the AI service
Vertical Slice	Complete implementation of a limited functional scope

Appendix B — Acronyms

- AI — Artificial Intelligence
 - API — Application Programming Interface
 - JSON — JavaScript Object Notation
 - OCR — Optical Character Recognition
 - PDF — Portable Document Format
 - RAG — Retrieval-Augmented Generation
 - LLM — Large Language Model
 - ONSSA — Office National de Sécurité Sanitaire des Produits Alimentaires
-

Appendix C — Main Deliverables

The project produces the following deliverables:

- Knowledge Ingestion Pipeline
- Knowledge Corpus
- Instruction Dataset
- Embedding Dataset
- Specialized ONSSA LLM (QLoRA Adapter)
- Vector Database
- Retrieval-Augmented Generation Service
- FastAPI REST API
- Docker Deployment Configuration
- Technical Documentation

- Evaluation Report

Appendix D — Main Technologies

Component	Technology
Backend API	FastAPI
LLM Runtime	Ollama
Fine-Tuning	QLoRA
Embeddings	Sentence Transformers (or selected embedding model)
Vector Database	FAISS / ChromaDB (final selection during implementation)
Containerization	Docker
Programming Language	Python

Appendix E — References

The design of this project is based on established concepts and practices in:

- Retrieval-Augmented Generation (RAG)
 - Parameter-Efficient Fine-Tuning (PEFT/QLoRA)
 - Knowledge Engineering
 - Semantic Search
 - Large Language Models
 - ONSSA regulatory documentation and Official Bulletins
-

End of Document

Ce format clôt le PRD de manière concise tout en restant professionnel. Tu obtiens ainsi un document complet couvrant la vision, l'architecture, les spécifications fonctionnelles et techniques, le déploiement, l'intégration, la feuille de route, les risques et les perspectives d'évolution, sans alourdir inutilement les derniers chapitres.